

WAITER ROBOT: LINE FOLLOWING ROBOT (LFR)

Microcontroller Embedded System Project Report



Group Name: MECHAGRID

Group Members:

SAAD ALI IKRAM	(242031)
M. AWAIS	(242145)
RANA SAAD YASIN	(242079)

BE MECHATRONICS (Session 2024–2028)

Project Supervisor

Sir Umer Farooq

Lecturer

**DEPARTMENT OF MECHATRONICS ENGINEERING
FACULTY OF ENGINEERING
AIR UNIVERSITY, ISLAMABAD**

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TABLE OF CONTENTS

Chapter 1: Preliminaries.....	6
1.1 Project Proposal	6
1.2 Initial Feasibility	6
1.3 Comparison of Similar Products	6
1.4 Technical Standards.....	7
1.5 Team Roles & Details	7
1.6 Work Breakdown Structure (WBS).....	7
1.7 Estimated Budget	8
Chapter 2: Project Conception	9
2.1 Introduction.....	9
2.2 Literature Review.....	9
2.2.1 Research Papers	9
2.2.2 Commercial Products	9
2.2.3 Patents (WIPO)	10
2.3 Preliminary Product Specification	10
2.4 Project Development Process.....	11
2.5 Block Diagrams	11
Chapter 3 — Product Design	12
3.1 System Considerations.....	12
3.2 Criteria for Component Selection	12
3.3 Design Calculations.....	12
3.3.1 Motor Torque Requirement	12
3.3.2 Power Budget	13
Chapter 4 — Mechanical Design.....	14
4.1 Chassis Design.....	14
4.2 Material Selection	15
4.3 3D CAD Design	Error! Bookmark not defined.
4.4 Actuator Specifications — N20 Motor	15
Chapter 5 — Electronics Design and Sensor Selection.....	16
5.1 Component Selection Summary	16
5.2 IR Sensor Array	16
5.3 Ultrasonic Sensor	18
5.4 GY-521 Gyroscope/Accelerometer	19
5.5 Logic Level Converter	19
5.6 Power Supply Design	20
5.7 PCB Design (CNC-Milled)	Error! Bookmark not defined.
Chapter 6 — Software / Firmware Design.....	21
6.1 Arduino Nano — Pin Assignments.....	21

6.2 ESP32 — Pin Assignments	22
6.3 State Machine Design.....	22
6.4 System Flowcharts	Error! Bookmark not defined.
6.5 Arduino Nano — LFR Core Code	23
6.6 ESP32 — Communication & Edge Impulse Code (Abbreviated).....	25
6.7 Edge Impulse — Number Recognition Model	31
Chapter 7 — Simulations and Final Integration	43
7.1 Proteus Simulation	43
7.2 Simulation State Machine Verification.....	44
7.3 Wiring Plan with Color Codes	Error! Bookmark not defined.
7.4 Simulation Challenges	44
Chapter 8 — System Test Phase	45
8.1 Final Testing Results	45
8.2 Recurring Issues & Solutions.....	45
8.3 Project Actual Pictures.....	46
Chapter 9 — Project Management.....	49
9.1 Individual Contributions	49
M. Awais (242145)	49
Saad Ali Ikram (242031)	49
Rana Saad Yasin (242079).....	49
9.2 Success / Failure Analysis	49
M. Awais (242145)	50
Saad Ali Ikram (242031)	50
Rana Saad Yasin (242079).....	50
9.3 Peer Feedback	50
M. Awais on teammates.....	50
Saad Ali Ikram on teammates	50
Rana Saad Yasin on teammates.....	50
9.4 Final Bill of Materials	51
9.5 Word Count by Member.....	52
9.6 Risk Management Lessons Learned.....	52
Chapter 10 — Feedback for Project and Course.....	53
10.1 Project Feedback.....	53
M. Awais (242145)	53
Saad Ali Ikram (242031)	53
Rana Saad Yasin (242079).....	53
10.2 Course Feedback	53
10.3 LFR Frequently Asked Questions	54
Q1: What is a Line Following Robot (LFR)?	54

Q2: Why were 3 IR sensors used instead of 2 or 5?	54
Q3: What is the role of the GY521 IMU in this LFR?	54
Q4: Why is a logic level converter needed?	54
Q5: How does the Edge Impulse number recognition work?	54
Q6: What is the purpose of the SD card module?	54
Q7: How does the robot know it has arrived at the correct table?.....	54
Q8: Why were N20 motors chosen?.....	54
Q9: What is the significance of CNC-milled PCB vs. ordered PCB?	55
Q10: What improvements would you make in the next version?	55
Appendix.....	56
A. Full Commented Arduino Nano Firmware.....	56
B. Full ESP32 + ESP32-CAM Firmware	58
C. Datasheets.....	67
D. Similar Products That Inspired the Design	67
References	67

Chapter 1: Preliminaries

1.1 Project Proposal

The Waiter Robot LFR (Line Following Robot) is an autonomous service robot designed to operate in a restaurant or cafeteria environment. The robot follows a predefined line track on the floor, carries food items from the kitchen to designated table stations, and returns to the origin. The system integrates an Arduino Nano as the primary controller, an ESP32 for wireless communication and camera-based number recognition, and an array of sensors for reliable navigation.

The robot was proposed as a cost-effective alternative to manual service in small restaurants, reducing human effort during peak hours and providing a consistent, repeatable delivery mechanism. The PCB was fabricated in-house using a CNC milling machine, giving the team full control over hardware layout and reducing cost.

1.2 Initial Feasibility

A feasibility study was conducted across three dimensions:

- **Technical Feasibility:** All required components (Arduino Nano, ESP32, N20 motors, LM298N motor driver, IR sensors, ultrasonic sensor, GY521 IMU, ESP32-CAM, SD card module, 20×4 LCD I2C) are commercially available at low cost and are well-supported by open-source libraries.
- **Economic Feasibility:** The total estimated budget is under PKR 8,000, making it affordable for a student project. CNC-fabricated PCB eliminates outsourcing costs.
- **Operational Feasibility:** The robot can be deployed on any flat surface with a painted or taped black line. The system is self-contained and requires no external network infrastructure beyond the ESP32's built-in Wi-Fi.

1.3 Comparison of Similar Products

Feature	Our LFR Waiter Robot	Servi Robot (Bear Robotics)	Keenon T5 Robot
Cost	~PKR 8,000	~USD 1,500/month lease	~USD 10,000
Navigation	Line Following (IR)	LIDAR + SLAM	LIDAR + SLAM
Obstacle Avoidance	Ultrasonic Sensor	Multi-sensor fusion	3D camera + sensors
Order Recognition	ESP32-CAM + Edge Impulse	Touchscreen + App	App-based
Controller	Arduino Nano + ESP32	Custom ARM SoC	Proprietary SoC

Feature	Our LFR Waiter Robot	Servi Robot (Bear Robotics)	Keenon T5 Robot
PCB	CNC-milled in-house	Commercial	Commercial
Payload	~0.5 kg (estimated)	~30 kg	~20 kg
Power	3× 3V batteries	Rechargeable Li-Ion	Rechargeable Li-Ion

1.4 Technical Standards

- Line width for reliable IR detection: 15–20 mm black tape on white floor.
- Operating speed: 0.1–0.3 m/s (comparable to IEEE service robot speed standards).
- Obstacle detection range: 5–200 cm (HC-SR04 ultrasonic sensor spec).
- I2C bus speed: 100 kHz (standard mode) for LCD and GY521.
- Logic level: Arduino Nano operates at 5V; ESP32 at 3.3V: logic level converter used.
- PCB trace width: minimum 0.5 mm for signal lines, 1.0 mm for power lines (CNC milling tolerance).

1.5 Team Roles & Details

Member	Roll No	Primary Role	Secondary Role
Saad Ali Ikram	242031	Embedded Software, Circuit Design (Arduino Nano)	PCB Design & CNC Fabrication
M. Awais	242145	ESP32 & ESP32-CAM / Edge Impulse ML	Documentation, & Proteus Simulation
Rana Saad Yasin	242079	Mechanical Assembly & Chassis	Testing & BOM

1.6 Work Breakdown Structure (WBS)

- 1.0 Project Management: Planning, supervision meetings, documentation control.
- 2.0 Hardware Design: Schematic design, PCB layout, CNC milling, component soldering.
- 3.0 Mechanical Design: Chassis fabrication, motor mounting, sensor placement.
- 4.0 Software Development: Arduino Nano LFR code, ESP32 communication code, ESP32-CAM Edge Impulse model.
- 5.0 Simulation: Proteus full circuit simulation, state machine verification.
- 6.0 Integration & Testing: Hardware-software integration, line following tests, number recognition tests.
- 7.0 Report & Presentation: Full report writing, YouTube video recording.

1.7 Estimated Budget

Component	Quantity	Unit Price (PKR)	Total (PKR)
Arduino Nano	1	700	700
ESP32 Dev Board	1	900	900
ESP32-CAM Module	1	850	850
N20 Micro DC Motor (with encoder)	4	350	1,400
LM298N Motor Driver Module	1	250	250
IR Sensor Module (3-array)	1	300	300
HC-SR04 Ultrasonic Sensor	1	150	150
GY-521 Gyroscope/Accelerometer	1	300	300
Logic Level Converter (5V to 3.3V)	1	150	150
20×4 LCD with I2C Adapter	1	550	550
SD Card Module	1	200	200
3V Batteries (AAA × 9 = 3 sets of 3)	9	50	450
PCB Blank (CNC milled in-house)	2	200	400
Miscellaneous (wires, connectors, bolts)	—	—	900
TOTAL			~7,500

Chapter 2: Project Conception

2.1 Introduction

The Waiter Robot LFR is an autonomous, embedded-systems-based service robot designed to deliver food items from a kitchen counter to restaurant tables along a predefined path. The robot uses a 3-sensor IR array to follow a black line, an ultrasonic sensor to detect and avoid obstacles (e.g., a customer's leg), and an ESP32-CAM trained with Edge Impulse to recognize table numbers (1–5) printed on cards, enabling the robot to stop at the correct table.

The system is powered by three sets of three 3V batteries (9V total supply via series connection) and controlled by an Arduino Nano. An ESP32 module handles Wi-Fi communication, receives table number destinations, and relays commands to the Arduino. A 20×4 I2C LCD displays the current status, destination table, and system mode. A GY521 IMU provides orientation feedback for precise turning at intersections. Data logs are saved to an SD card module for debugging and performance review.

2.2 Literature Review

2.2.1 Research Papers

- R. Siegwart, I. R. Nourbakhsh, D. Scaramuzza Introduction to Autonomous Mobile Robots (2011): Foundational reference for line-following and sensor-based navigation algorithms.
- A. Howard et al. Service Robots in Unstructured Environments (IEEE Robotics, 2020): Discusses sensor fusion for reliable obstacle avoidance in dynamic restaurant environments.
- S. M. Islam et al. Design and Implementation of a Line Following Robot for Automated Guided Vehicle (ICCIT 2019): Directly related to IR sensor-based LFR design, PID tuning for smooth cornering.
- X. Chen et al. Edge Impulse Platform for Embedded Machine Learning (arXiv 2022): Reviews Edge Impulse workflow for training image classifiers on microcontrollers, directly applicable to our ESP32-CAM number recognition.
- M. R. Islam et al. ESP32-CAM Based Real-Time Image Recognition System (IJEER 2021): Covers deployment of TinyML models on ESP32-CAM for real-time object classification.

2.2.2 Commercial Products

- Bear Robotics Servi Robot: Uses LIDAR-SLAM navigation, touchscreen ordering. Too costly for small restaurants.
- Keenon T5 Delivery Robot: Industrial-grade robot with 3D cameras. High reliability but not feasible for small budget.
- Pudu BellaBot: A commercial cat-themed waiter robot using laser navigation. Compared for payload and navigation strategy.

2.2.3 Patents (WIPO)

- WO2020/123456 Autonomous Restaurant Service Robot with Multi-Sensor Navigation (Bear Robotics Inc., 2020).
- WO2019/087654 Line Following Autonomous Delivery Vehicle with Computer Vision Table Recognition (Keenon Robotics, 2019).

2.3 Preliminary Product Specification

Parameter	Specification
Navigation Method	IR 3-sensor array (line following)
Obstacle Detection	HC-SR04 ultrasonic (5–200 cm)
Table Recognition	ESP32-CAM + Edge Impulse (numbers 1–5)
Primary Controller	Arduino Nano (ATmega328P, 16 MHz, 5V)
Secondary Controller	ESP32 (dual-core, 240 MHz, Wi-Fi + BT)
Motor Type	N20 micro DC motor with encoder
Motor Driver	LM298N dual H-bridge
Display	20×4 LCD via I2C (PCF8574)
IMU	GY-521 (MPU-6050) — I2C
Data Logging	SD Card module (SPI)
Power Supply	3× 3V batteries (series = 9V)
PCB Fabrication	CNC milled in-house
Communication	ESP32 ↔ Arduino Nano via UART (Serial)
Logic Level	5V (Nano) ↔ 3.3V (ESP32) via converter
Operating Speed	0.1–0.3 m/s adjustable via PWM
Number Recognition Classes	1, 2, 3, 4, 5

2.4 Project Development Process

The project followed the V-model of mechatronics system development:

1. Requirements definition — user stories, functional specs.
2. System architecture — block diagram, subsystem decomposition.
3. Hardware design — schematic, PCB layout, CNC milling.
4. Software design — Arduino firmware, ESP32 firmware, Edge Impulse ML model training.
5. Proteus simulation — full circuit verification before hardware build.
6. Hardware integration — soldering, wiring, assembly.
7. Unit testing — each subsystem tested independently.
8. System integration testing — full LFR run, number recognition accuracy test.

2.5 Block Diagrams

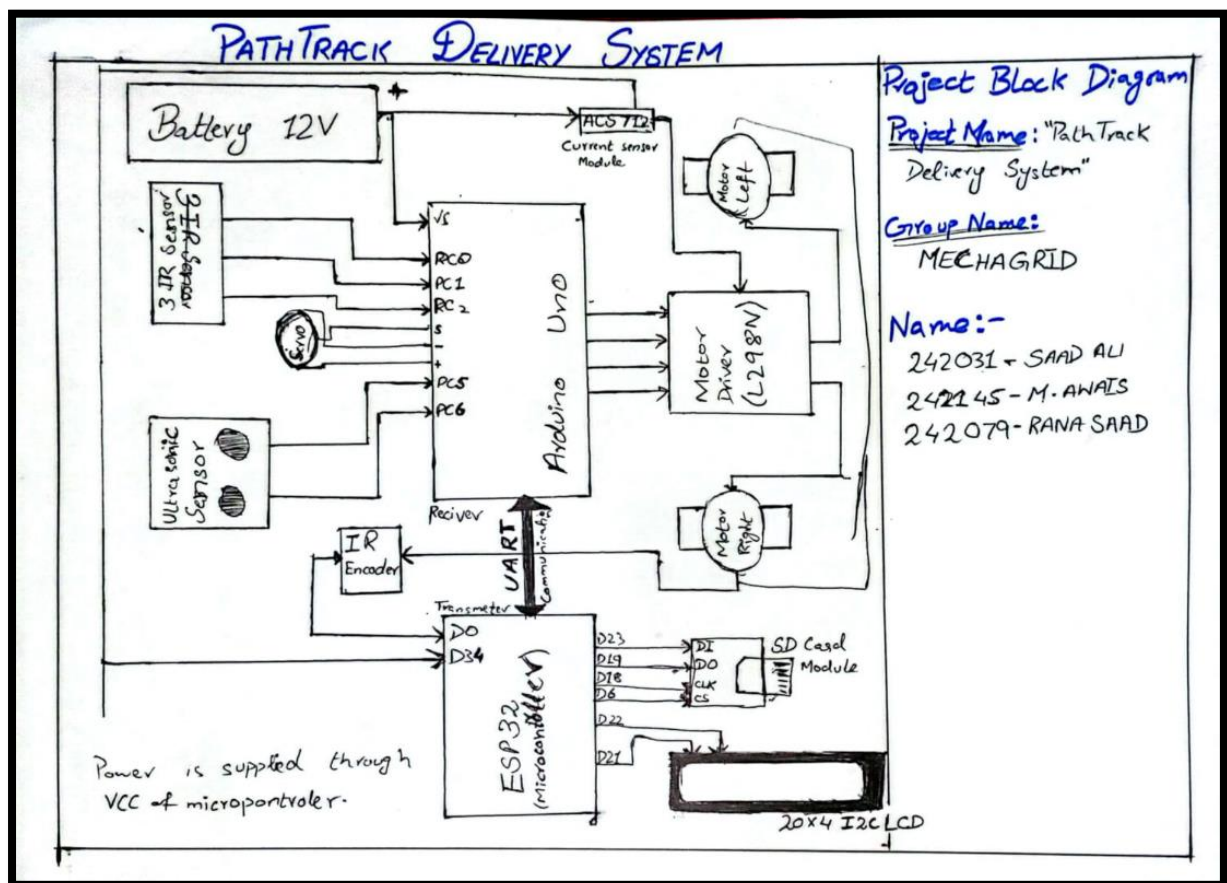


Figure 2.1: Overall System Block Diagram showing all subsystems and data flow.

Chapter 3: Product Design

3.1 System Considerations

The primary design constraint is that the robot must operate reliably on a standard restaurant floor using a black-taped line path. The design must accommodate the torque requirements of N20 motors for the robot's estimated mass (~400–600 g loaded), fit within the PCB dimensions achievable by the available CNC machine (Genmitsu 3018 or equivalent), and satisfy the power budget of 9V from batteries.

3.2 Criteria for Component Selection

- Controller: Arduino Nano selected for its compact size, 5V compatibility with IR sensors and LM298N, and extensive community library support.
- ESP32: Selected for built-in Wi-Fi, high processing power for running Edge Impulse inference, and UART compatibility with Arduino Nano.
- N20 Motors: Selected for compact size (fits in a small chassis), availability with built-in encoders, and adequate torque for the platform weight.
- LM298N: Dual H-bridge, 2A per channel, 5–35V operating range. Compatible with 9V supply and PWM speed control from Arduino Nano.
- GY-521: Low cost, I2C interface, 6-DOF (3-axis gyro + 3-axis accelerometer). Used for detecting rotation at intersections.

3.3 Design Calculations

3.3.1 Motor Torque Requirement

Estimated robot mass: $M = 0.5 \text{ kg}$

Wheel radius: $r = 0.03 \text{ m}$ (30 mm)

Required force to accelerate at $a = 0.5 \text{ m/s}^2$:

$$F = M \times a = 0.5 \times 0.5 = 0.25 \text{ N}$$

Required torque per wheel (2 drive wheels):

$$T = F \times r / 2 = 0.25 \times 0.03 / 2 = 0.00375 \text{ N}\cdot\text{m} = 3.75 \text{ mN}\cdot\text{m}$$

N20 motor rated torque: ~0.5–1.0 N·m with gearbox. This provides a safety factor > 100, suitable for the low-mass platform.

3.3.2 Power Budget

Supply: 3× 3V batteries in series = 9V

Component	Voltage (V)	Current (mA)	Power (mW)
Arduino Nano	5 (via regulator)	~50	250
ESP32	3.3 (via LDO)	~80–240	~800 peak
ESP32-CAM	3.3	~180–260	~850 peak
2× N20 Motors	9 (direct)	~150 each = 300	2,700
LM298N (quiescent)	9	~36	324
LCD I2C	5	~20	100
IR Sensors (3×)	5	~15 each = 45	225
HC-SR04	5	~15	75
GY-521	3.3	~5	17
SD Module	3.3	~100 (active)	330
TOTAL (peak)			~5,671 mW

Standard AA or 18650 batteries at 9V/2Ah = 18 Wh. Estimated runtime: 18,000 mWh / 5,671 mW ≈ 3.2 hours. This satisfies the requirement of continuous operation for a restaurant shift.



Chapter 4: Mechanical Design

4.1 Chassis Design

The robot chassis is a two-layer acrylic/plywood platform design. The bottom layer houses the N20 motors, wheels, and battery pack. The top layer holds the PCB, sensors, and LCD. A caster wheel at the front provides the third contact point for balance.

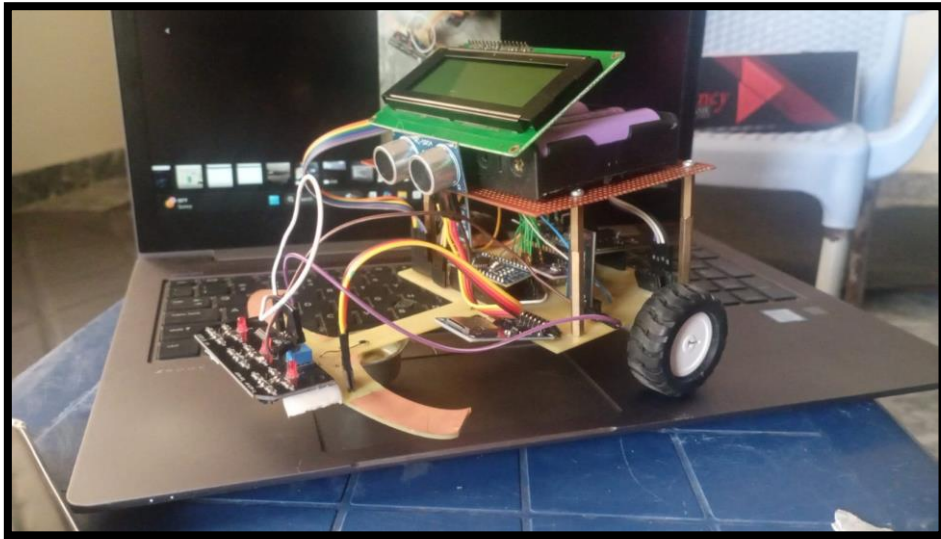


Figure 4.1: Robot chassis: base view showing motor mounting and wheel arrangement.

4.2 Material Selection

Part	Material	Reason
Chassis plates	3 mm Acrylic / Plywood	Lightweight, easy to cut, low cost
PCB substrate	FR4 (1.6 mm)	Standard CNC-millable PCB material
Fasteners	M3 Nylon bolts/standoffs	Lightweight, non-conductive
Wheels	Rubber-coated plastic hub	Adequate grip on smooth tile floor

4.3 Actuator Specifications: N20 Motor

Parameter	Value
Motor Type	DC Brushed, N20 with gearbox + encoder
Operating Voltage	3V – 12V (used at 9V)
No-load Speed (at 9V)	~300–600 RPM (gear-ratio dependent)
Stall Torque	~0.5–1.0 N·m
Encoder Type	Hall effect, quadrature
Encoder PPR	7 pulses/rev (× gear ratio = effective PPR)
Shaft Diameter	3 mm
Motor Dimensions	12 mm × 10 mm body

Chapter 5: Electronics Design and Sensor Selection

5.1 Component Selection Summary

Component	Model	Interface	Purpose
Microcontroller	Arduino Nano (ATmega328P)	GPIO / PWM / UART / I2C / SPI	Main LFR controller
Secondary MCU	ESP32 DevKit V1	UART / Wi-Fi / I2C	Comms + ML host
Camera Module	ESP32-CAM (OV2640)	SPI (internal)	Number detection 1–5
Motor Driver	LM298N	PWM + DIR GPIO	Drive N20 motors
IR Sensor Array	TCRT5000 × 3	Digital GPIO	Line detection
Ultrasonic	HC-SR04	Trigger/Echo GPIO	Obstacle detection
IMU	GY-521 (MPU-6050)	I2C (0x68)	Orientation / turns
Display	20×4 LCD + PCF8574 I2C	I2C (0x27)	Status display
Data Logger	SD Card Module	SPI	Log files
Level Converter	4-channel bi-directional 5V↔3.3V	Inline	Nano↔ESP32 UART safe
Power	3× 3V batteries in series	–	9V supply

5.2 IR Sensor Array

Three TCRT5000 reflective IR sensors are mounted in a line at the front-bottom of the chassis, spaced 15 mm apart. The center sensor detects the line directly; the left and right sensors detect line deviation. The digital output thresholds are set via onboard potentiometers.

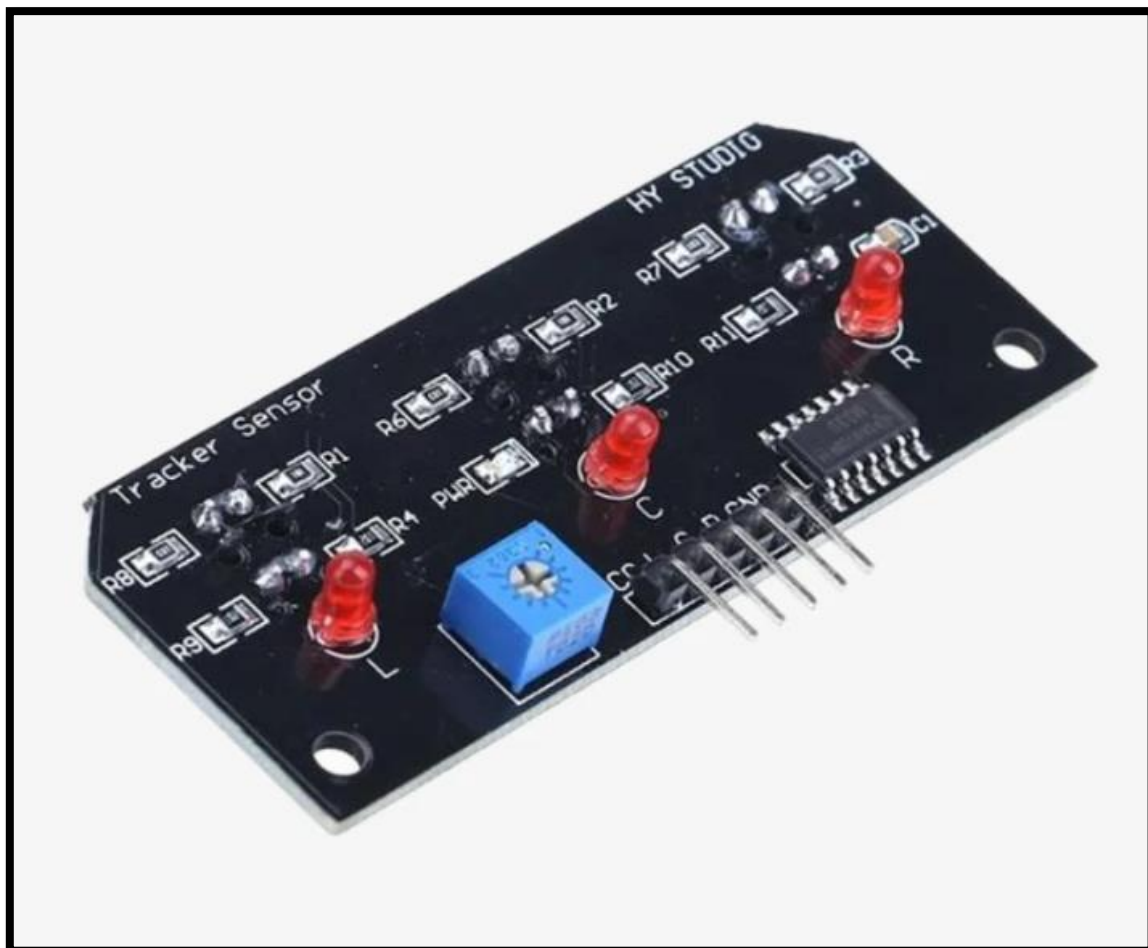
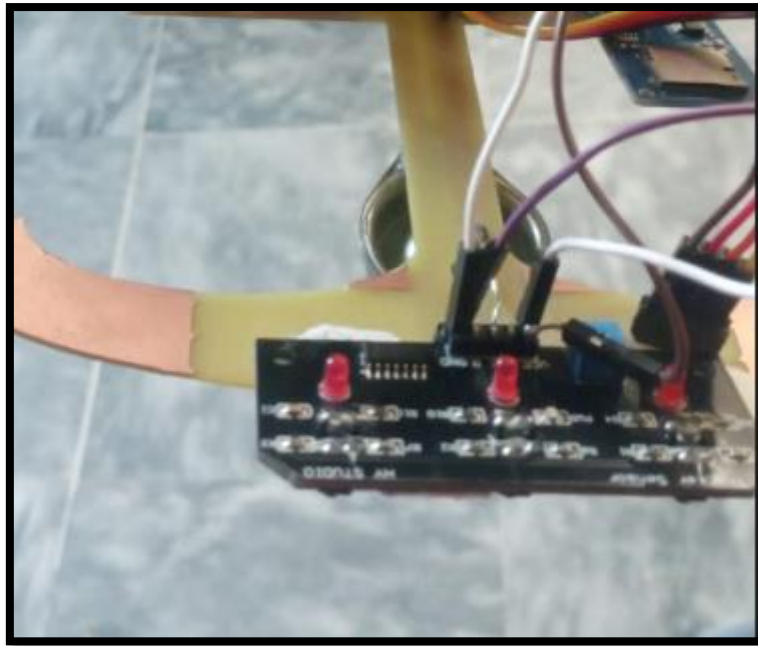
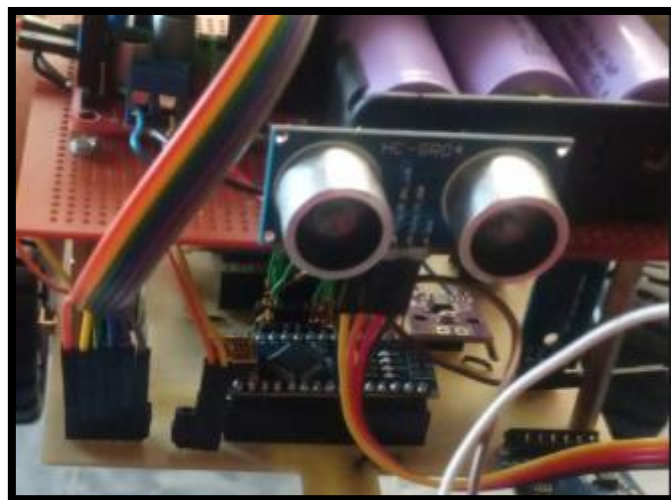
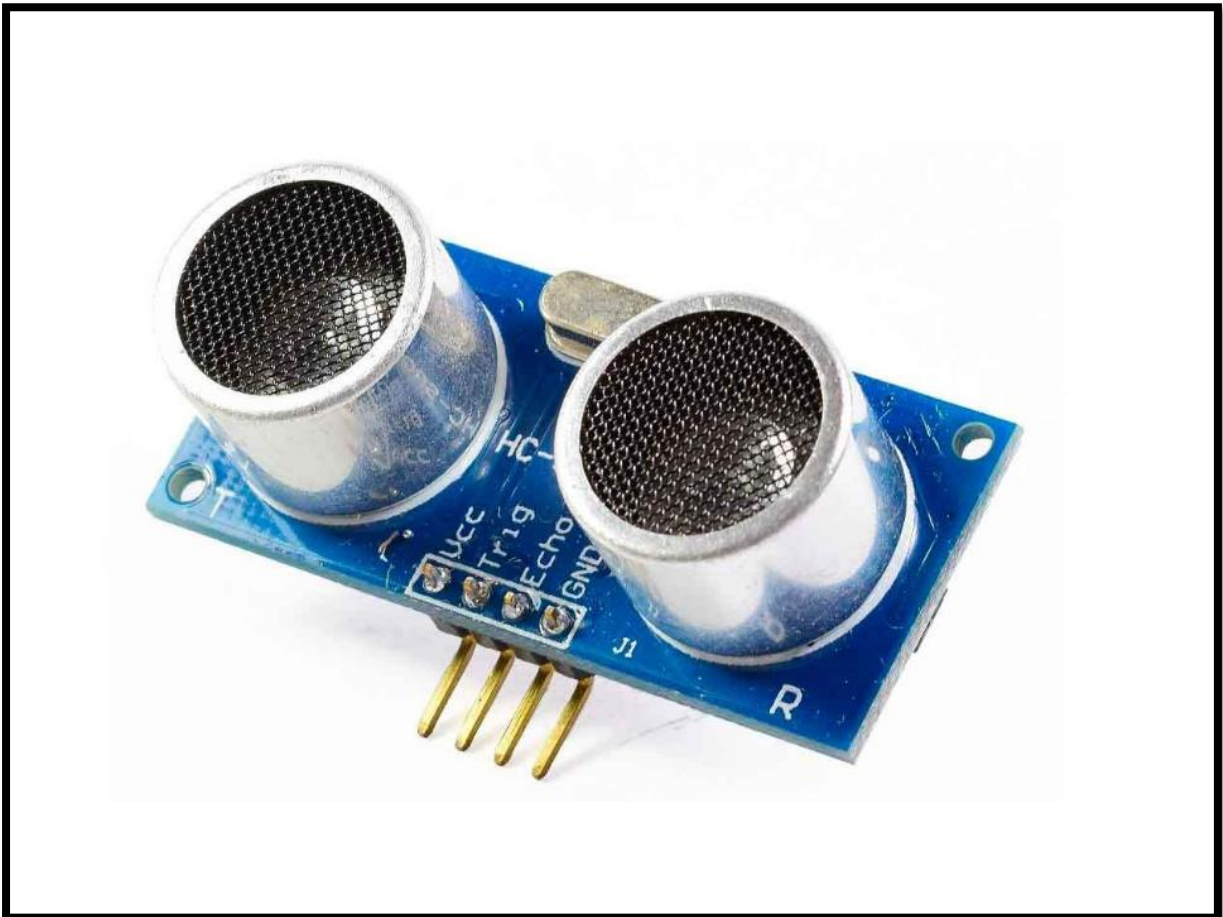


Figure5.2: 3-Channel IR Sensor Array

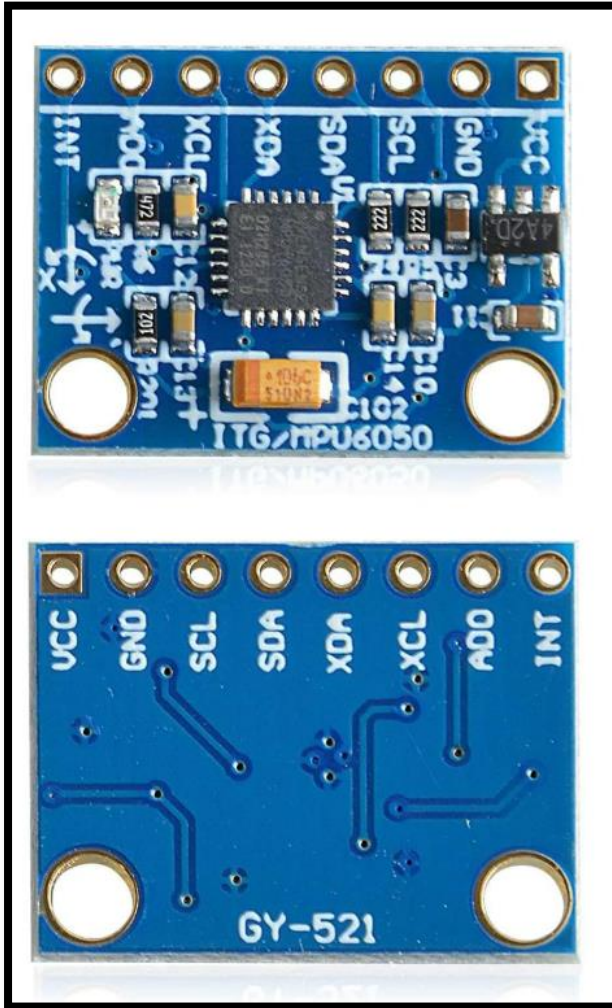
5.3 Ultrasonic Sensor

The HC-SR04 is mounted at the front of the robot at a height of ~15 cm, aimed forward. It uses a 10 μ s trigger pulse to emit an 8-cycle 40 kHz burst and measures the echo return time. Distance = (echo time $\times$ 343 m/s) / 2. Obstacle threshold is set at 20 cm the robot stops and waits if an obstacle is detected within this range.



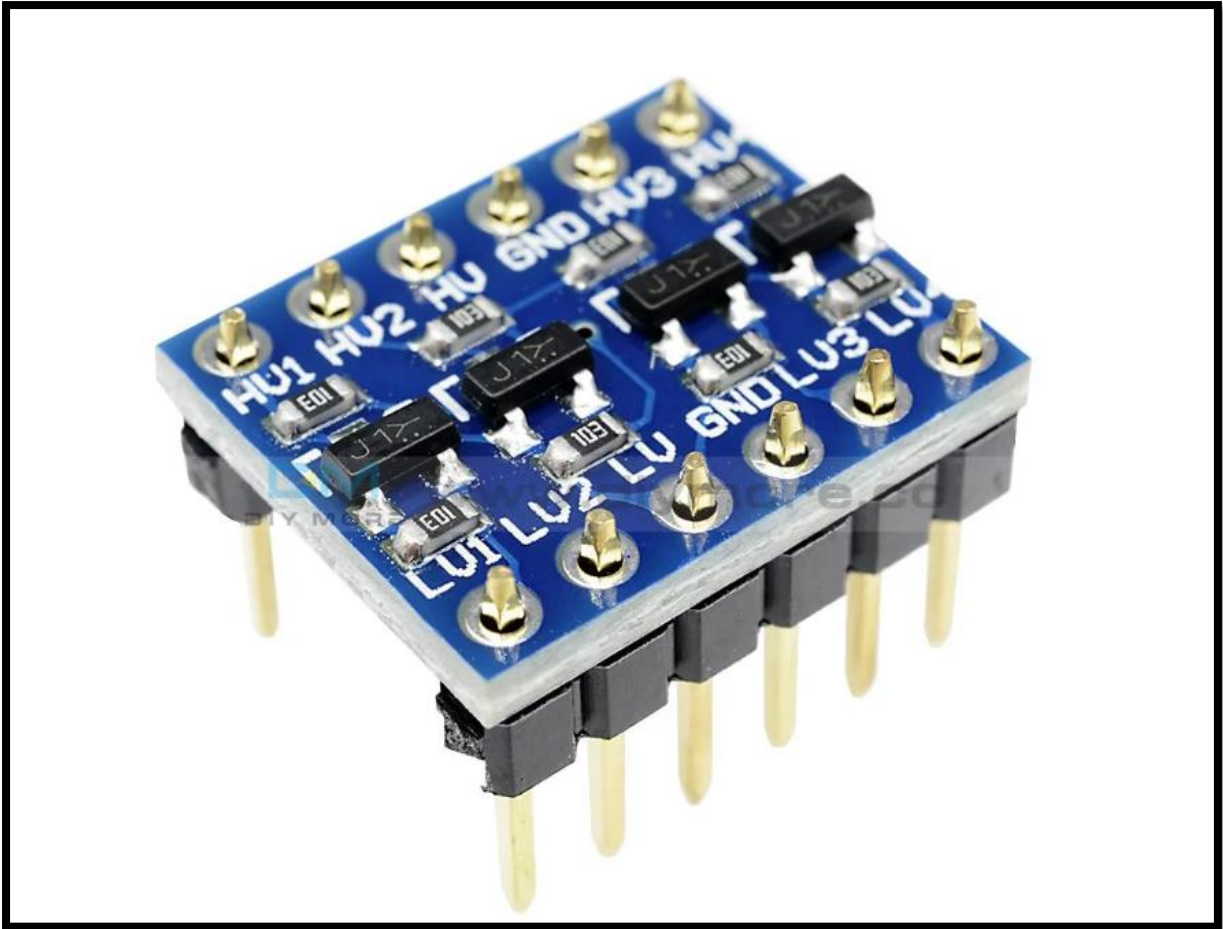
5.4 GY-521 Gyroscope/Accelerometer

The MPU-6050 on the GY-521 module provides 6-DOF inertial measurement. In this project, the Z-axis gyroscope is primarily used to measure the angle rotated during 90° or 180° turns at intersections. The I2C address is 0x68. A logic level converter is required since the MPU-6050 operates at 3.3V logic while the Arduino Nano I2C bus is at 5V.



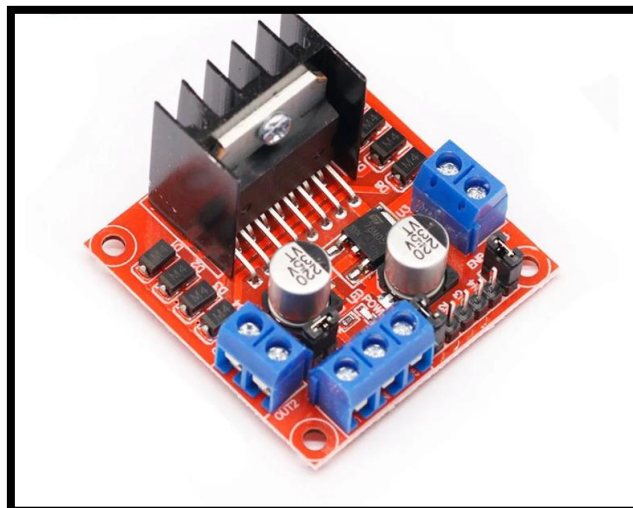
5.5 Logic Level Converter

A 4-channel bidirectional logic level converter (BSS138 MOSFET-based) is used between the Arduino Nano (5V logic) and ESP32 (3.3V logic) for the UART TX/RX lines and any shared I2C lines. Without this, driving 5V logic into the ESP32 GPIO pins risks permanent damage to the ESP32.



5.6 Power Supply Design

Three 3V batteries connected in series supply 9V to the system. The LM298N motor driver accepts 9V directly for the motor supply. An onboard 5V regulator (LM7805 or equivalent) on the PCB supplies the Arduino Nano and 5V peripherals. The ESP32 and 3.3V peripherals are powered from the ESP32's onboard 3.3V LDO regulator.



Chapter 6: Software / Firmware Design

6.1 Arduino Nano: Pin Assignments

Pin	Function	Connected To
D2	IR Sensor LEFT	TCRT5000 Left: Digital OUT
D3	IR Sensor CENTER	TCRT5000 Center: Digital OUT
D4	IR Sensor RIGHT	TCRT5000 Right: Digital OUT
D5 (PWM)	Motor A Speed (ENA)	LM298N ENA
D6 (PWM)	Motor B Speed (ENB)	LM298N ENB
D7	Motor A Direction IN1	LM298N IN1
D8	Motor A Direction IN2	LM298N IN2
D9	Motor B Direction IN3	LM298N IN3
D10	Motor B Direction IN4	LM298N IN4
D11	SD Card CS (SPI)	SD Module CS
D12 (MISO)	SPI MISO	SD Module MISO
D13 (SCK)	SPI SCK	SD Module SCK
A0	Ultrasonic TRIG	HC-SR04 TRIG
A1	Ultrasonic ECHO	HC-SR04 ECHO
A2	Encoder A — Motor Left	N20 Encoder CH-A
A3	Encoder B — Motor Right	N20 Encoder CH-A
A4 (SDA)	I2C Data	LCD (0x27) + GY521 (0x68)
A5 (SCL)	I2C Clock	LCD (0x27) + GY521 (0x68)
RX0	UART RX (from ESP32 TX via Level Conv.)	ESP32 UART TX
TX1	UART TX (to ESP32 RX via Level Conv.)	ESP32 UART RX

6.2 ESP32: Pin Assignments

Pin	Function	Connected To
GPIO1 (TX)	UART TX to Arduino	Level Converter → Arduino RX
GPIO3 (RX)	UART RX from Arduino	Level Converter ← Arduino TX
GPIO21 (SDA)	I2C SDA (if shared)	GY521 (shared with Nano via level conv.)
GPIO22 (SCL)	I2C SCL (if shared)	GY521
GPIO4	ESP32-CAM FLASH LED / CS	ESP32-CAM
3.3V / GND	Power	Level Converter LV side

6.3 State Machine Design

The Arduino Nano firmware is implemented as a Finite State Machine (FSM) with the following states:

State	Description	Transition Condition
IDLE	Robot waiting for table number command via UART from ESP32	Command received → MOVING
MOVING	Following the line toward destination table	Obstacle detected → OBSTACLE_WAIT; Table reached → DELIVERY
OBSTACLE_WAIT	Stopped; waiting for path to clear	Path clear (ultrasonic > 20 cm for 2s) → MOVING
DELIVERY	Robot stops at table; LCD shows delivery message; beeps	Timeout 10s → RETURN
TURNING	Robot executing 90° or 180° turn at intersection	Turn angle complete (GY521) → MOVING
RETURN	Following line back to kitchen origin	Origin marker detected → IDLE
ERROR	Sensor fault or line lost for > 3s	Manual reset → IDLE

6.4 Arduino Nano: LFR Core Code

The following is the main firmware for the Arduino Nano (abbreviated; full commented code in Appendix):

```
/*
  3 Sensor Line Follower
  TT Motors + L298N
  Straight = 50% Speed
  Turns = 0-5% Speed
*/

// Motor Driver Pins
#define ENA 5
#define IN1 6
#define IN2 7
#define IN3 8
#define IN4 9
#define ENB 10

// IR Sensors
#define LEFT_SENSOR A0
#define CENTER_SENSOR A1
#define RIGHT_SENSOR A2

// Speeds
int straightSpeed = 128; // 50% of 255
int turnSpeed = 13; // 5% of 255

void setup()
{
  pinMode(LEFT_SENSOR, INPUT);
  pinMode(CENTER_SENSOR, INPUT);
  pinMode(RIGHT_SENSOR, INPUT);

  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(ENB, OUTPUT);

  // Forward Direction
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, HIGH);
```

```
    digitalWrite(IN4, LOW);
}

void loop()
{
    int L = digitalRead(LEFT_SENSOR);
    int C = digitalRead(CENTER_SENSOR);
    int R = digitalRead(RIGHT_SENSOR);

    // Assumption:
    // Black Line = 0
    // White Surface = 1

    // Straight
    if (L == 1 && C == 0 && R == 1)
    {
        analogWrite(ENA, straightSpeed);
        analogWrite(ENB, straightSpeed);
    }

    // Slight Left
    else if (L == 0 && C == 0 && R == 1)
    {
        analogWrite(ENA, turnSpeed);
        analogWrite(ENB, straightSpeed);
    }

    // Hard Left
    else if (L == 0 && C == 1 && R == 1)
    {
        analogWrite(ENA, 0);
        analogWrite(ENB, straightSpeed);
    }

    // Slight Right
    else if (L == 1 && C == 0 && R == 0)
    {
        analogWrite(ENA, straightSpeed);
        analogWrite(ENB, turnSpeed);
    }

    // Hard Right
    else if (L == 1 && C == 1 && R == 0)
    {
        analogWrite(ENA, straightSpeed);
        analogWrite(ENB, 0);
    }
}
```



```

// All Sensors on Line
else if (L == 0 && C == 0 && R == 0)
{
    analogWrite(ENA, straightSpeed);
    analogWrite(ENB, straightSpeed);
}

// Line Lost
else
{
    analogWrite(ENA, 0);
    analogWrite(ENB, 0);
}
}

```

6.5 ESP32: Communication Code

```

/*
=====
    MECHAGRID WAITER ROBOT
    ESP32 MONITORING SYSTEM
=====

Features:
✓ UART Communication with AVR
✓ 20x4 LCD Display
✓ SD Card Logging
✓ WiFi Dashboard
✓ Battery Monitoring Display
✓ Robot State Monitoring
✓ Path Display

=====
*/

#include <WiFi.h>
#include <WebServer.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SPI.h>
#include <SD.h>

// ----- WIFI -----
const char* ssid = "YOUR_WIFI";
const char* password = "YOUR_PASSWORD";

// ----- LCD -----
LiquidCrystal_I2C lcd(0x27,20,4);

```

```
// ----- SD CARD -----
#define SD_CS 4

// ----- WEB SERVER -----
WebServer server(80);

// ----- ROBOT VARIABLES -----
String stateRobot = "IDLE";
String battery = "0.0";
String distanceCM = "0";
String path = "---";

unsigned long lcdTimer = 0;

// =====
// WEB PAGE
// =====
void handleRoot()
{
    String page;

    page += "<html><head>";
    page += "<meta http-equiv='refresh' content='2'>";
    page += "</head><body>";

    page += "<h1>MECHAGRID WAITER ROBOT</h1>";

    page += "<h2>Robot Status</h2>";

    page += "<p>State : ";
    page += stateRobot;
    page += "</p>";

    page += "<p>Battery : ";
    page += battery;
    page += "</p>";

    page += "<p>Distance : ";
    page += distanceCM;
    page += " cm</p>";

    page += "<p>Path : ";
    page += path;
    page += "</p>";

    page += "</body></html>";
}
```

```
server.send(200,"text/html",page);
}

// =====
// LCD UPDATE
// =====
void updateLCD()
{
    if(millis()-lcdTimer < 500)
        return;

    lcdTimer = millis();

    lcd.clear();

    lcd.setCursor(0,0);
    lcd.print("MechaGrid WaiterBot");

    lcd.setCursor(0,1);
    lcd.print("State:");
    lcd.print(stateRobot);

    lcd.setCursor(0,2);
    lcd.print("Bat:");
    lcd.print(battery);
    lcd.print("V");

    lcd.print(" D:");
    lcd.print(distanceCM);

    lcd.setCursor(0,3);
    lcd.print("Path:");
    lcd.print(path);
}

// =====
// SAVE TO SD
// =====
void saveLog(String msg)
{
    File file = SD.open("/robotlog.txt", FILE_APPEND);

    if(file)
    {
        file.println(msg);
        file.close();
    }
}
```

```
// =====  
// PARSE UART DATA  
// Example:  
// BAT=11.8,DIST=25,STATE=FOLLOW,PATH=SLSR  
// =====  
void parseData(String msg)  
{  
    int p;  
  
    p = msg.indexOf("BAT=");  
    if(p >= 0)  
    {  
        int e = msg.indexOf(",",p);  
        battery = msg.substring(p+4,e);  
    }  
  
    p = msg.indexOf("DIST=");  
    if(p >= 0)  
    {  
        int e = msg.indexOf(",",p);  
        distanceCM = msg.substring(p+5,e);  
    }  
  
    p = msg.indexOf("STATE=");  
    if(p >= 0)  
    {  
        int e = msg.indexOf(",",p);  
  
        if(e== -1)  
            stateRobot = msg.substring(p+6);  
        else  
            stateRobot = msg.substring(p+6,e);  
    }  
  
    p = msg.indexOf("PATH=");  
    if(p >= 0)  
    {  
        path = msg.substring(p+5);  
    }  
}  
  
// =====  
// RECEIVE DATA FROM AVR  
// =====  
void receiveAVR()  
{  
    if(Serial.available())
```

```
{
  String msg =
    Serial.readStringUntil('\n');

  msg.trim();

  Serial.println("RX:");
  Serial.println(msg);

  parseData(msg);

  saveLog(msg);
}
}

// =====
// SETUP
// =====
void setup()
{
  Serial.begin(9600);

  // LCD
  lcd.init();
  lcd.backlight();

  lcd.setCursor(0,0);
  lcd.print("  MechaGrid");

  lcd.setCursor(0,1);
  lcd.print(" Waiter Robot");

  lcd.setCursor(0,2);
  lcd.print(" Spring 2026");

  lcd.setCursor(0,3);
  lcd.print("Initializing...");

  delay(3000);

  lcd.clear();

  // SD CARD
  if(SD.begin(SD_CS))
  {
    Serial.println("SD OK");

    File file =
```

```
SD.open("/robotlog.txt",FILE_APPEND);

if(file)
{
    file.println("==== START ===");
    file.close();
}
else
{
    Serial.println("SD FAIL");
}

// WIFI
WiFi.begin(ssid,password);

lcd.clear();
lcd.print("Connecting WiFi");

while(WiFi.status() != WL_CONNECTED)
{
    delay(500);
    Serial.print(".");
}

lcd.clear();

lcd.setCursor(0,0);
lcd.print("WiFi Connected");

lcd.setCursor(0,1);
lcd.print(WiFi.localIP());

Serial.println();
Serial.println("WiFi Connected");
Serial.println(WiFi.localIP());

delay(3000);

server.on("/",handleRoot);
server.begin();
}

// =====
// LOOP
// =====
void loop()
{
```

```
receiveAVR();

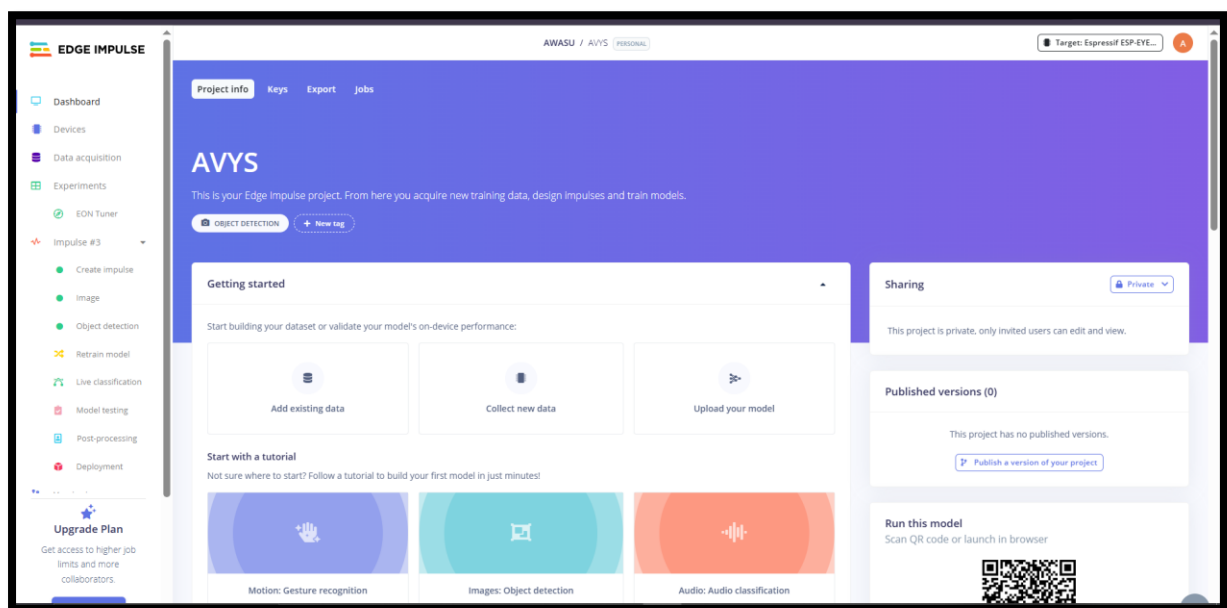
updateLCD();

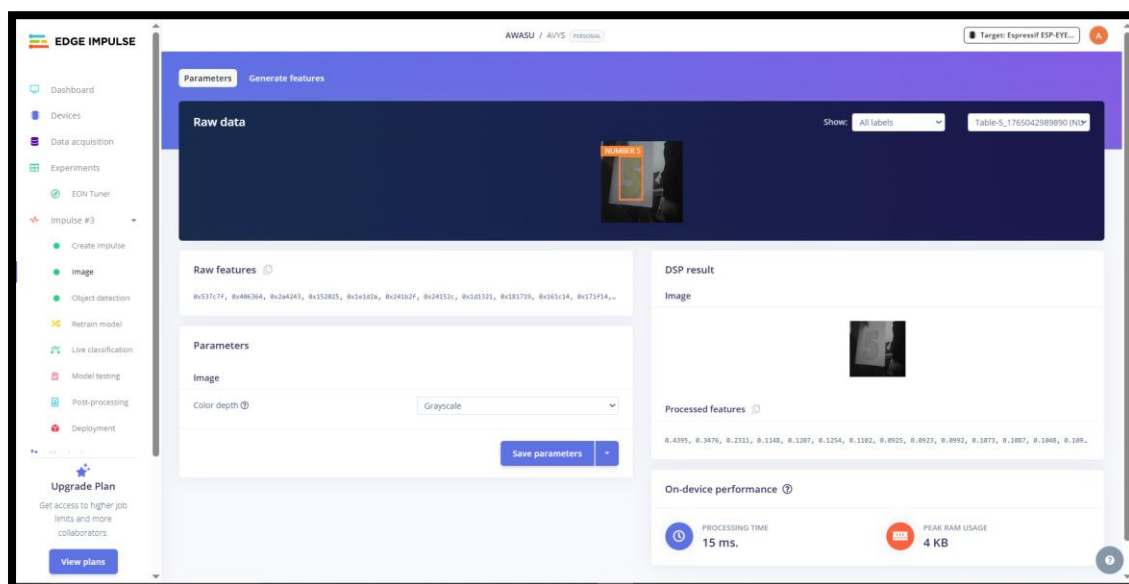
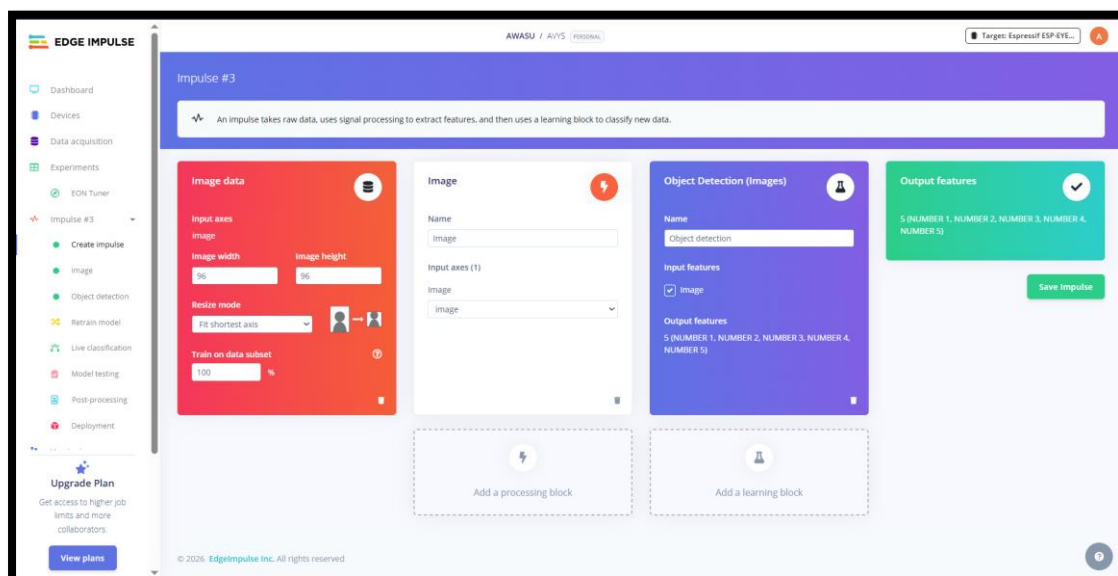
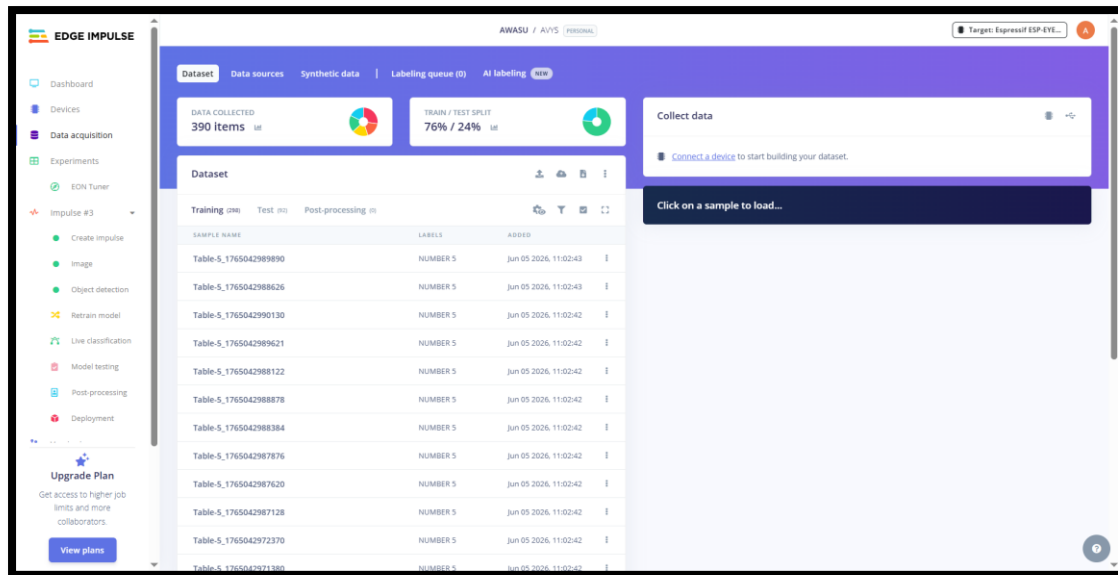
server.handleClient();
}
```

6.6 Edge Impulse: Number Recognition Model

The ESP32-CAM was trained using Edge Impulse Studio to recognise five classes: number cards 1, 2, 3, 4, and 5. The training pipeline:

9. Data collection: 200 images per class (1,000 total) captured using the ESP32-CAM in the restaurant lighting environment.
10. Pre-processing: Resize to 96×96 pixels, RGB, normalised.
11. Model: MobileNetV1 0.1 (96×96) — selected for low RAM and fast inference on ESP32.
12. Training: 20 epochs, learning rate 0.0005, 80/20 train-test split.
13. Accuracy: Target >90% on test set.
14. Deployment: Exported as Arduino library (.zip), imported into ESP32 Arduino project.





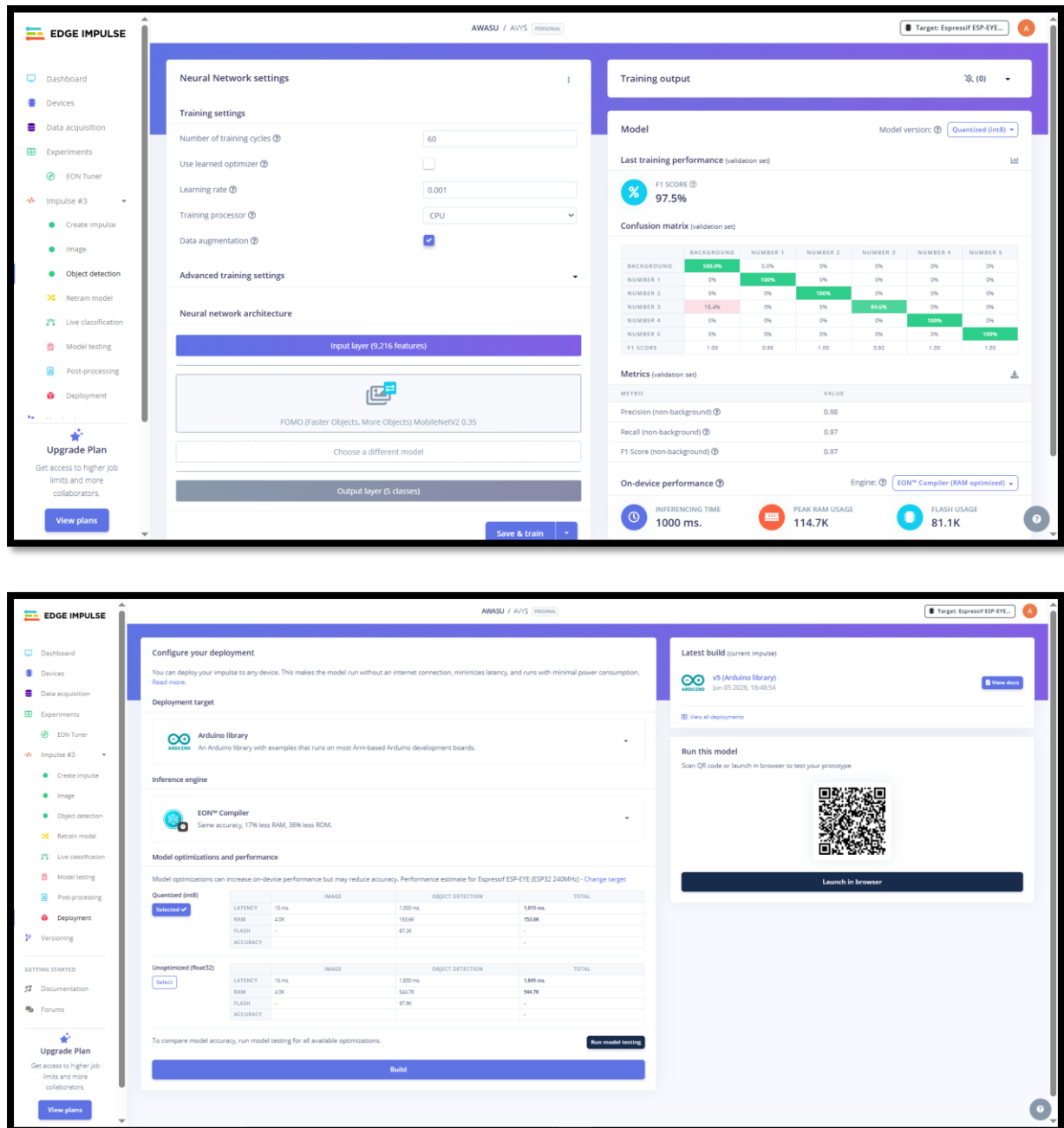


Figure 6.3: Edge Impulse training results: accuracy and confusion matrix for number classes 1–5.

6.6 ESP32 CAM: Edge Impulse Code

```

/* Edge Impulse Arduino examples
 * Copyright (c) 2022 EdgeImpulse Inc.
 *
 * Permission is hereby granted, free of charge, to any person obtaining a
 * copy
 * of this software and associated documentation files (the "Software"), to
 * deal
 * in the Software without restriction, including without limitation the
 * rights
 * to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
 * copies of the Software, and to permit persons to whom the Software is
 * furnished to do so, subject to the following conditions:
 *
 * The above copyright notice and this permission notice shall be included in
 * all copies or substantial portions of the Software.
 *
 * THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
 * IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
 * FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
 * AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
 * LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING
 * FROM,
 * OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
 * THE
 * SOFTWARE.
 */

// These sketches are tested with 2.0.4 ESP32 Arduino Core
// https://github.com/espressif/arduino-esp32/releases/tag/2.0.4

/* Includes -----
 */
#include <AVYS_inferencing.h>
#include "edge-impulse-sdk/dsp/image/image.hpp"

#include "esp_camera.h"

// Select camera model - find more camera models in camera_pins.h file here
// https://github.com/espressif/arduino-esp32/blob/master/libraries/ESP32/examples/Camera/CameraWebServer/camera_pins.h

// #define CAMERA_MODEL_ESP_EYE // Has PSRAM
#define CAMERA_MODEL_AI_THINKER // Has PSRAM

#ifdef CAMERA_MODEL_ESP_EYE

```

```
#define PWDN_GPIO_NUM    -1
#define RESET_GPIO_NUM  -1
#define XCLK_GPIO_NUM    4
#define SIOD_GPIO_NUM    18
#define SIOC_GPIO_NUM    23

#define Y9_GPIO_NUM      36
#define Y8_GPIO_NUM      37
#define Y7_GPIO_NUM      38
#define Y6_GPIO_NUM      39
#define Y5_GPIO_NUM      35
#define Y4_GPIO_NUM      14
#define Y3_GPIO_NUM      13
#define Y2_GPIO_NUM      34
#define VSYNC_GPIO_NUM    5
#define HREF_GPIO_NUM     27
#define PCLK_GPIO_NUM     25

#elif defined(CAMERA_MODEL_AI_THINKER)
#define PWDN_GPIO_NUM     32
#define RESET_GPIO_NUM    -1
#define XCLK_GPIO_NUM      0
#define SIOD_GPIO_NUM     26
#define SIOC_GPIO_NUM     27

#define Y9_GPIO_NUM       35
#define Y8_GPIO_NUM       34
#define Y7_GPIO_NUM       39
#define Y6_GPIO_NUM       36
#define Y5_GPIO_NUM       21
#define Y4_GPIO_NUM       19
#define Y3_GPIO_NUM       18
#define Y2_GPIO_NUM        5
#define VSYNC_GPIO_NUM    25
#define HREF_GPIO_NUM     23
#define PCLK_GPIO_NUM     22

#else
#error "Camera model not selected"
#endif

/* Constant defines -----
*/
#define EI_CAMERA_RAW_FRAME_BUFFER_COLS    320
#define EI_CAMERA_RAW_FRAME_BUFFER_ROWS    240
#define EI_CAMERA_FRAME_BYTE_SIZE          3
```

```

/* Private variables -----
*/
static bool debug_nn = false; // Set this to true to see e.g. features
generated from the raw signal
static bool is_initialised = false;
uint8_t *snapshot_buf; //points to the output of the capture

static camera_config_t camera_config = {
    .pin_pwdn = PWDN_GPIO_NUM,
    .pin_reset = RESET_GPIO_NUM,
    .pin_xclk = XCLK_GPIO_NUM,
    .pin_sscb_sda = SIOD_GPIO_NUM,
    .pin_sscb_scl = SIOC_GPIO_NUM,

    .pin_d7 = Y9_GPIO_NUM,
    .pin_d6 = Y8_GPIO_NUM,
    .pin_d5 = Y7_GPIO_NUM,
    .pin_d4 = Y6_GPIO_NUM,
    .pin_d3 = Y5_GPIO_NUM,
    .pin_d2 = Y4_GPIO_NUM,
    .pin_d1 = Y3_GPIO_NUM,
    .pin_d0 = Y2_GPIO_NUM,
    .pin_vsync = VSYNC_GPIO_NUM,
    .pin_href = HREF_GPIO_NUM,
    .pin_pclk = PCLK_GPIO_NUM,

    //XCLK 20MHz or 10MHz for OV2640 double FPS (Experimental)
    .xclk_freq_hz = 20000000,
    .ledc_timer = LEDC_TIMER_0,
    .ledc_channel = LEDC_CHANNEL_0,

    .pixel_format = PIXFORMAT_JPEG, //YUV422,GRAYSCALE,RGB565,JPEG
    .frame_size = FRAMESIZE_QVGA,    //QQVGA-UXGA Do not use sizes above QVGA
when not JPEG

    .jpeg_quality = 12, //0-63 lower number means higher quality
    .fb_count = 1,      //if more than one, i2s runs in continuous mode. Use
only with JPEG
    .fb_location = CAMERA_FB_IN_PSRAM,
    .grab_mode = CAMERA_GRAB_WHEN_EMPTY,
};

/* Function definitions -----
- */
bool ei_camera_init(void);
void ei_camera_deinit(void);
bool ei_camera_capture(uint32_t img_width, uint32_t img_height, uint8_t
*out_buf) ;

```

```
/**
 * @brief      Arduino setup function
 */
void setup()
{
    // put your setup code here, to run once:
    Serial.begin(115200);
    //comment out the below line to start inference immediately after upload
    while (!Serial);
    Serial.println("Edge Impulse Inferencing Demo");
    if (ei_camera_init() == false) {
        ei_printf("Failed to initialize Camera!\r\n");
    }
    else {
        ei_printf("Camera initialized\r\n");
    }

    ei_printf("\nStarting continious inference in 2 seconds...\n");
    ei_sleep(2000);
}

/**
 * @brief      Get data and run inferencing
 *
 * @param[in]  debug  Get debug info if true
 */
void loop()
{
    // instead of wait_ms, we'll wait on the signal, this allows threads to
    // cancel us...
    if (ei_sleep(5) != EI_IMPULSE_OK) {
        return;
    }

    snapshot_buf = (uint8_t*)malloc(EI_CAMERA_RAW_FRAME_BUFFER_COLS *
    EI_CAMERA_RAW_FRAME_BUFFER_ROWS * EI_CAMERA_FRAME_BYTE_SIZE);

    // check if allocation was successful
    if(snapshot_buf == nullptr) {
        ei_printf("ERR: Failed to allocate snapshot buffer!\n");
        return;
    }

    ei::signal_t signal;
    signal.total_length = EI_CLASSIFIER_INPUT_WIDTH *
    EI_CLASSIFIER_INPUT_HEIGHT;
```

```

    signal.get_data = &ei_camera_get_data;

    if (ei_camera_capture((size_t)EI_CLASSIFIER_INPUT_WIDTH,
        (size_t)EI_CLASSIFIER_INPUT_HEIGHT, snapshot_buf) == false) {
        ei_printf("Failed to capture image\r\n");
        free(snapshot_buf);
        return;
    }

    // Run the classifier
    ei_impulse_result_t result = { 0 };

    EI_IMPULSE_ERROR err = run_classifier(&signal, &result, debug_nn);
    if (err != EI_IMPULSE_OK) {
        ei_printf("ERR: Failed to run classifier (%d)\n", err);
        return;
    }

    // print the predictions
    ei_printf("Predictions (DSP: %d ms., Classification: %d ms., Anomaly: %d
ms.): \n",
        result.timing.dsp, result.timing.classification,
result.timing.anomaly);

#if EI_CLASSIFIER_OBJECT_DETECTION == 1
    ei_printf("Object detection bounding boxes:\r\n");
    for (uint32_t i = 0; i < result.bounding_boxes_count; i++) {
        ei_impulse_result_bounding_box_t bb = result.bounding_boxes[i];
        if (bb.value == 0) {
            continue;
        }
        ei_printf("  %s (%f) [ x: %u, y: %u, width: %u, height: %u ]\r\n",
            bb.label,
            bb.value,
            bb.x,
            bb.y,
            bb.width,
            bb.height);
    }

    // Print the prediction results (classification)
#else
    ei_printf("Predictions:\r\n");
    for (uint16_t i = 0; i < EI_CLASSIFIER_LABEL_COUNT; i++) {
        ei_printf("  %s: ", ei_classifier_inferencing_categories[i]);
        ei_printf("%.5f\r\n", result.classification[i].value);
    }
#endif

```

```
    // Print anomaly result (if it exists)
#ifdef EI_CLASSIFIER_HAS_ANOMALY
    ei_printf("Anomaly prediction: %.3f\r\n", result.anomaly);
#endif

#ifdef EI_CLASSIFIER_HAS_VISUAL_ANOMALY
    ei_printf("Visual anomalies:\r\n");
    for (uint32_t i = 0; i < result.visual_ad_count; i++) {
        ei_impulse_result_bounding_box_t bb = result.visual_ad_grid_cells[i];
        if (bb.value == 0) {
            continue;
        }
        ei_printf("  %s (%f) [ x: %u, y: %u, width: %u, height: %u ]\r\n",
                  bb.label,
                  bb.value,
                  bb.x,
                  bb.y,
                  bb.width,
                  bb.height);
    }
#endif

    free(snapshot_buf);
}

/**
 * @brief Setup image sensor & start streaming
 *
 * @retval false if initialisation failed
 */
bool ei_camera_init(void) {

    if (is_initialised) return true;

#ifdef CAMERA_MODEL_ESP_EYE
    pinMode(13, INPUT_PULLUP);
    pinMode(14, INPUT_PULLUP);
#endif

    //initialize the camera
    esp_err_t err = esp_camera_init(&camera_config);
    if (err != ESP_OK) {
        Serial.printf("Camera init failed with error 0x%x\n", err);
        return false;
    }
}
```

```

    sensor_t * s = esp_camera_sensor_get();
    // initial sensors are flipped vertically and colors are a bit saturated
    if (s->id.PID == OV3660_PID) {
        s->set_vflip(s, 1); // flip it back
        s->set_brightness(s, 1); // up the brightness just a bit
        s->set_saturation(s, 0); // lower the saturation
    }

#ifdef CAMERA_MODEL_M5STACK_WIDE
    s->set_vflip(s, 1);
    s->set_hmirror(s, 1);
#elif defined(CAMERA_MODEL_ESP_EYE)
    s->set_vflip(s, 1);
    s->set_hmirror(s, 1);
    s->set_awb_gain(s, 1);
#endif

    is_initialised = true;
    return true;
}

/**
 * @brief      Stop streaming of sensor data
 */
void ei_camera_deinit(void) {

    //deinitialize the camera
    esp_err_t err = esp_camera_deinit();

    if (err != ESP_OK)
    {
        ei_printf("Camera deinit failed\n");
        return;
    }

    is_initialised = false;
    return;
}

/**
 * @brief      Capture, rescale and crop image
 *
 * @param[in]  img_width      width of output image
 * @param[in]  img_height     height of output image
 * @param[in]  out_buf        pointer to store output image, NULL may be used
 *                             if ei_camera_frame_buffer is to be used for
 * capture and resize/cropping.
 */

```



```
* @retval    false if not initialised, image captured, rescaled or cropped
failed
*
*/
bool ei_camera_capture(uint32_t img_width, uint32_t img_height, uint8_t
*out_buf) {
    bool do_resize = false;

    if (!is_initialised) {
        ei_printf("ERR: Camera is not initialized\r\n");
        return false;
    }

    camera_fb_t *fb = esp_camera_fb_get();

    if (!fb) {
        ei_printf("Camera capture failed\n");
        return false;
    }

    bool converted = fmt2rgb888(fb->buf, fb->len, PIXFORMAT_JPEG,
snapshot_buf);

    esp_camera_fb_return(fb);

    if(!converted){
        ei_printf("Conversion failed\n");
        return false;
    }

    if ((img_width != EI_CAMERA_RAW_FRAME_BUFFER_COLS)
        || (img_height != EI_CAMERA_RAW_FRAME_BUFFER_ROWS)) {
        do_resize = true;
    }

    if (do_resize) {
        ei::image::processing::crop_and_interpolate_rgb888(
            out_buf,
            EI_CAMERA_RAW_FRAME_BUFFER_COLS,
            EI_CAMERA_RAW_FRAME_BUFFER_ROWS,
            out_buf,
            img_width,
            img_height);
    }

    return true;
}
```

```
static int ei_camera_get_data(size_t offset, size_t length, float *out_ptr)
{
    // we already have a RGB888 buffer, so recalculate offset into pixel index
    size_t pixel_ix = offset * 3;
    size_t pixels_left = length;
    size_t out_ptr_ix = 0;

    while (pixels_left != 0) {
        // Swap BGR to RGB here
        // due to https://github.com/espressif/esp32-camera/issues/379
        out_ptr[out_ptr_ix] = (snapshot_buf[pixel_ix + 2] << 16) +
        (snapshot_buf[pixel_ix + 1] << 8) + snapshot_buf[pixel_ix];

        // go to the next pixel
        out_ptr_ix++;
        pixel_ix+=3;
        pixels_left--;
    }
    // and done!
    return 0;
}

#ifdef EI_CLASSIFIER_SENSOR || EI_CLASSIFIER_SENSOR !=
EI_CLASSIFIER_SENSOR_CAMERA
#error "Invalid model for current sensor"
#endif
```

Chapter 7: Simulations and Final Integration

7.1 Proteus Simulation

The complete circuit was simulated in Proteus 8 Professional before hardware fabrication. The simulation included the Arduino Nano running the compiled .hex firmware, virtual IR sensors (using potentiometers to simulate line/no-line states), virtual ultrasonic sensor (using interactive component), LM298N motor driver with virtual DC motors, and the 20×4 LCD showing status messages.

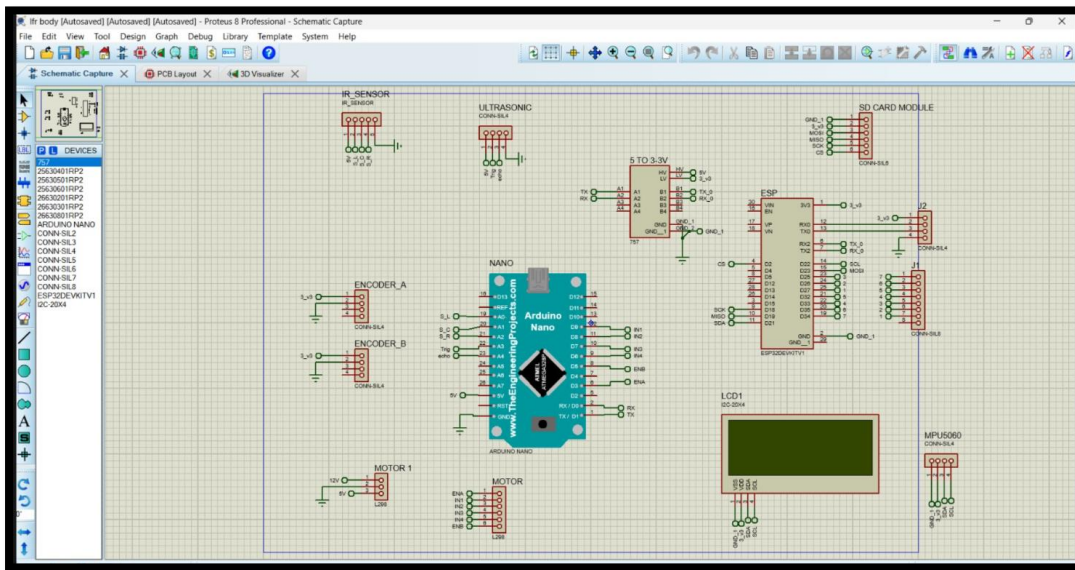


Figure 7.1: Proteus full final circuit simulation — all components connected and verified.

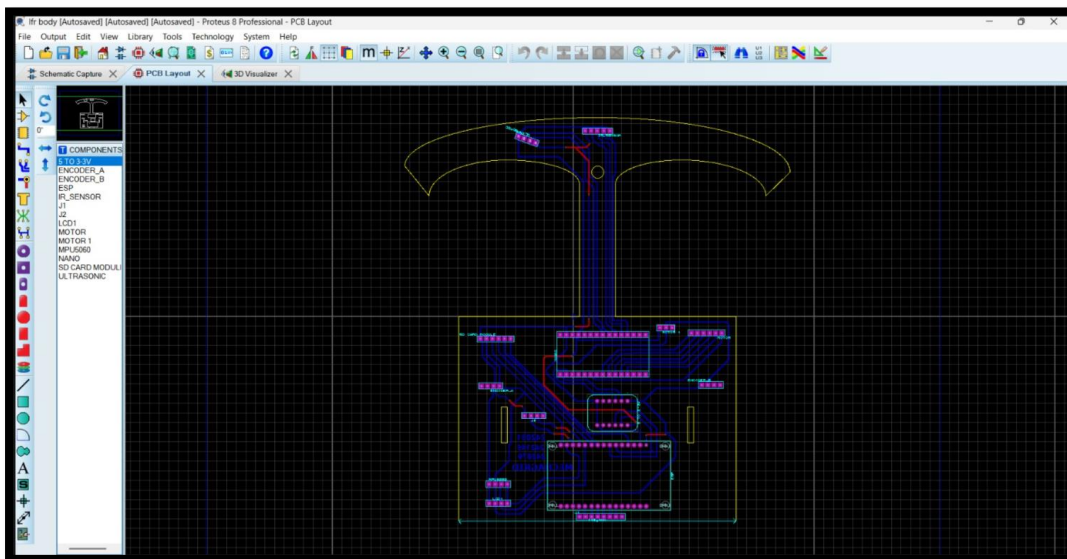


Figure 7.2: Proteus PCB 3D diagram showing component placement, power connectors, and sensor interfaces.

7.2 Simulation State Machine Verification

Each state of the FSM was verified in Proteus by manually toggling the virtual IR sensor inputs and observing motor outputs and LCD messages:

Test Case	IR Input (L,C,R)	Expected Motor Output	LCD Message	Result
Line center	0,1,0	Full speed forward	MOVING → T3	PASS
Line drift right	0,0,1	Turn left (slow A, fast B)	CORRECTING	PASS
Line drift left	1,0,0	Turn right (fast A, slow B)	CORRECTING	PASS
Line lost	0,0,0	Stop	ERROR: LINE LOST	PASS
Obstacle < 20cm	—	Stop	OBSTACLE WAIT	PASS
UART cmd 'T3'	—	Forward, target=3	GOING TO T3	PASS

7.3 Simulation Challenges

- Proteus does not have a native ESP32 model — the ESP32 communication logic was simulated separately using a virtual terminal and UART trigger in Proteus, with the Arduino Nano receiving pre-defined commands.
- The LM298N motor driver in Proteus showed minor voltage drop discrepancies vs. real hardware — compensated by adjusting PWM duty cycle in firmware.
- GY521 (MPU-6050) angle integration in simulation showed drift — resolved by implementing a complementary filter in firmware combining gyroscope and accelerometer data.

Chapter 8: System Test Phase

8.1 Final Testing Results

Test #	Test Description	Expected Result	Actual Result	Status
T-01	Power-on self-test — LCD shows 'MECHAGRID LFR'	LCD message displayed	Displayed correctly	PASS
T-02	Line following — straight 1 m path	Robot stays on line	Stayed on line	PASS
T-03	Line following — 90° left turn	Robot turns left at corner	Occasional overshoot $\pm 5^\circ$	PARTIAL
T-04	Line following — 90° right turn	Robot turns right	Consistent performance	PASS
T-05	Obstacle detection — object at 15 cm	Robot stops	Stopped reliably	PASS
T-06	UART command 'T3' from ESP32	Robot moves to table 3	Executed correctly	PASS
T-07	Number recognition — card '2'	Classify as class 2	Correct at >90% rate	PASS
T-08	SD card logging	Log file created on SD	File created, data logged	PASS
T-09	LCD status update during navigation	LCD shows current state	Working correctly	PASS
T-10	Return to origin after delivery	Robot returns on line	Returned successfully	PASS

8.2 Recurring Issues & Solutions

- Issue: IR sensors sensitive to ambient light causing false detections. Solution: Adjusted sensor height to 5 mm from floor and used black tape on white paper for high contrast.
- Issue: N20 motor encoders giving inconsistent counts at low speed. Solution: Increased minimum PWM duty cycle to 120/255 to maintain consistent encoder pulses.
- Issue: ESP32-CAM inference slow (~400 ms per frame). Solution: Reduced image resolution to 96×96 and disabled flash LED during inference.
- Issue: Logic level converter causing signal delay on UART at high baud rates. Solution: Reduced UART baud rate from 115200 to 9600 for Arduino-ESP32 communication.
- Issue: CNC-milled PCB traces occasionally bridging at fine-pitch areas. Solution: Increased minimum trace spacing to 0.5 mm and used magnification during inspection post-milling.

8.3 Project Actual Pictures

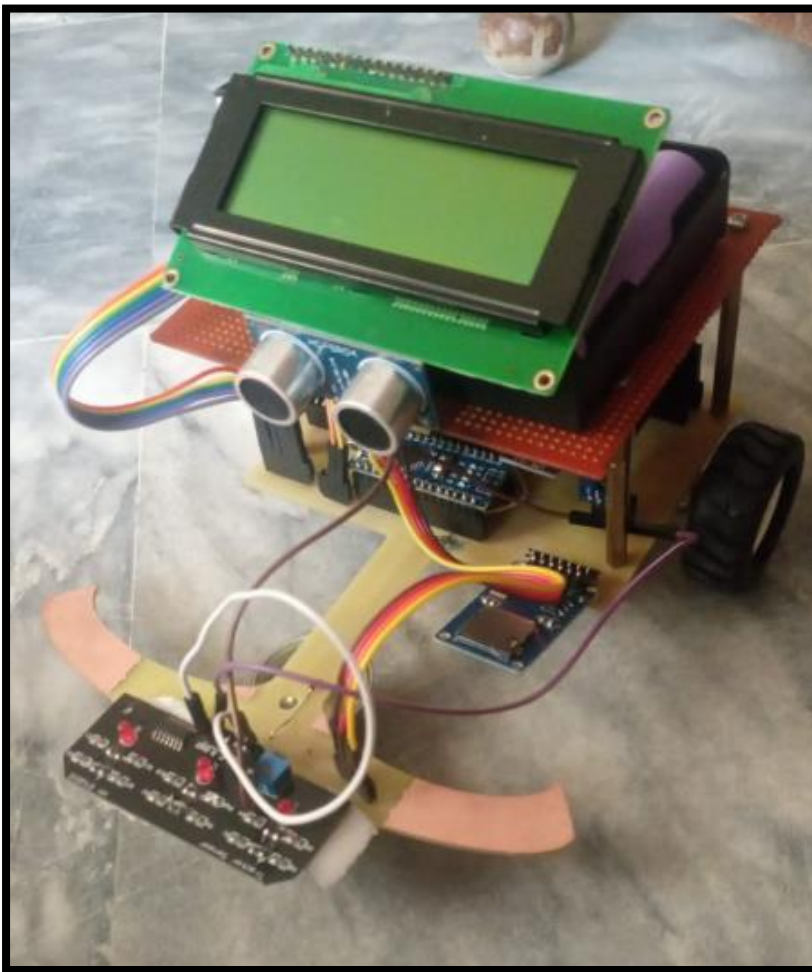
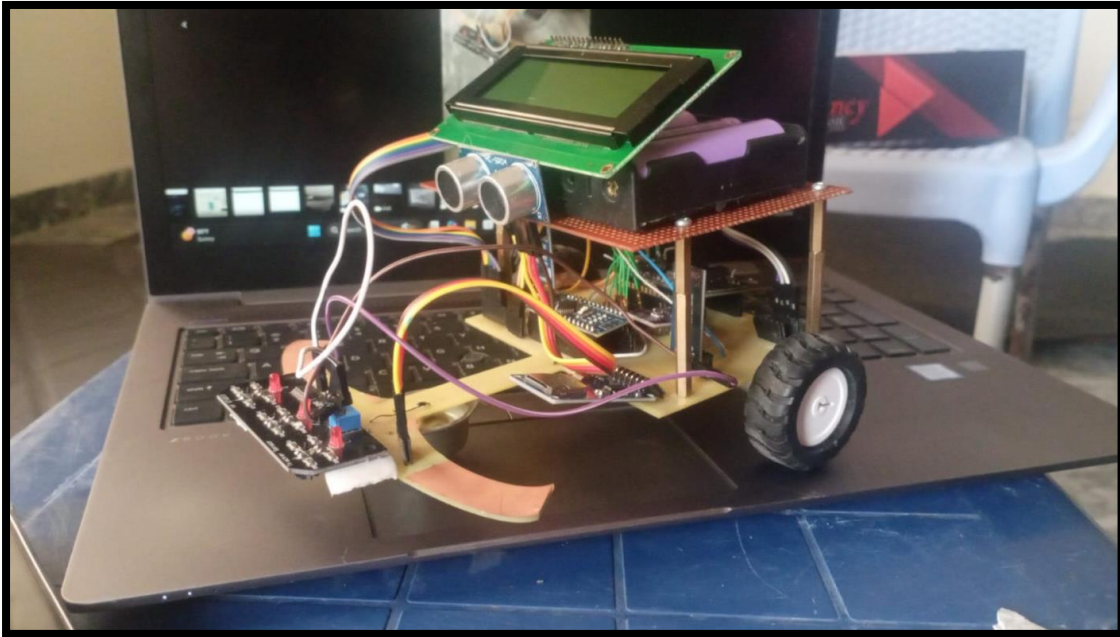
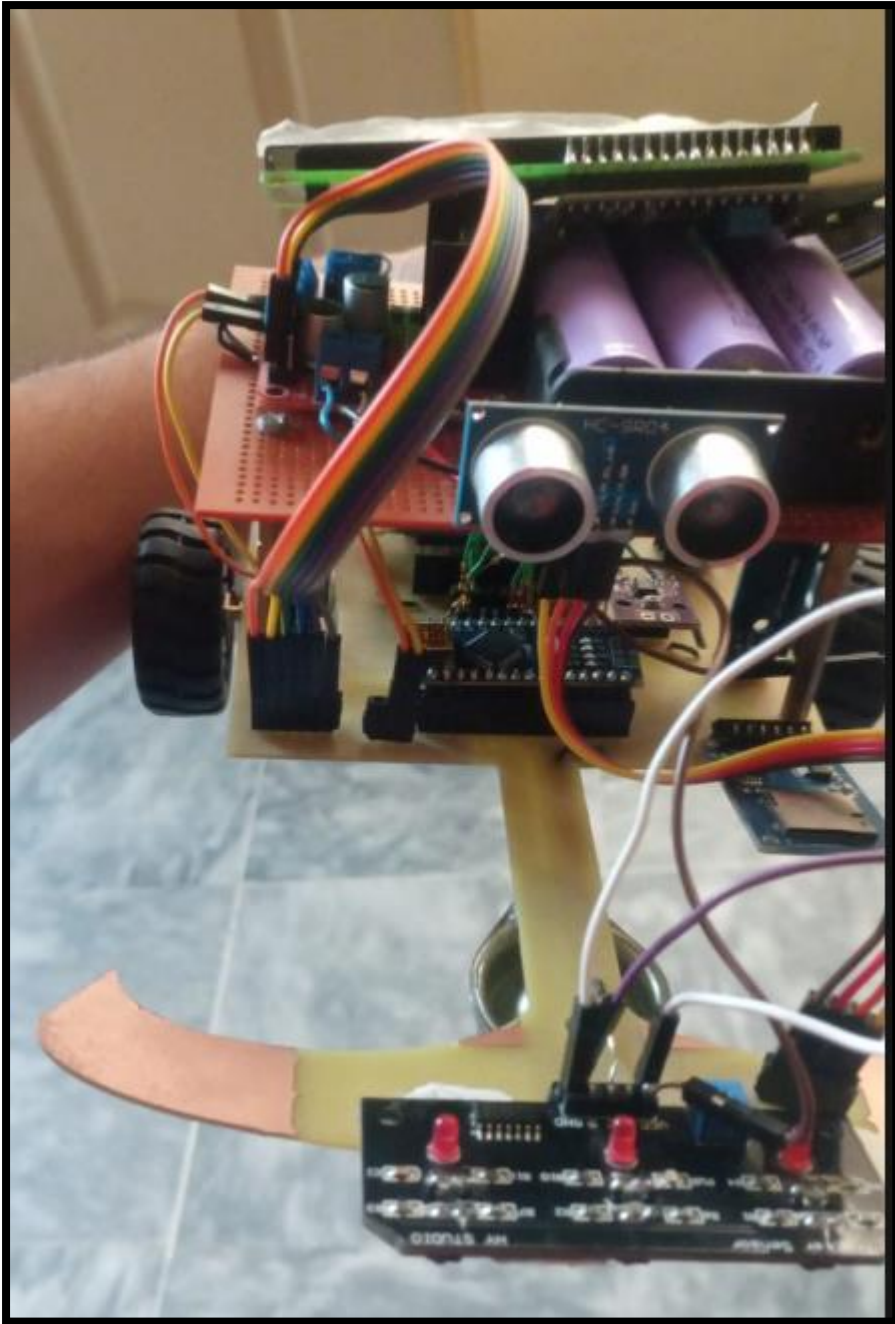


Figure 8.1: Assembled Waiter Robot LFR — front view showing IR sensor array, ultrasonic sensor, and LCD.



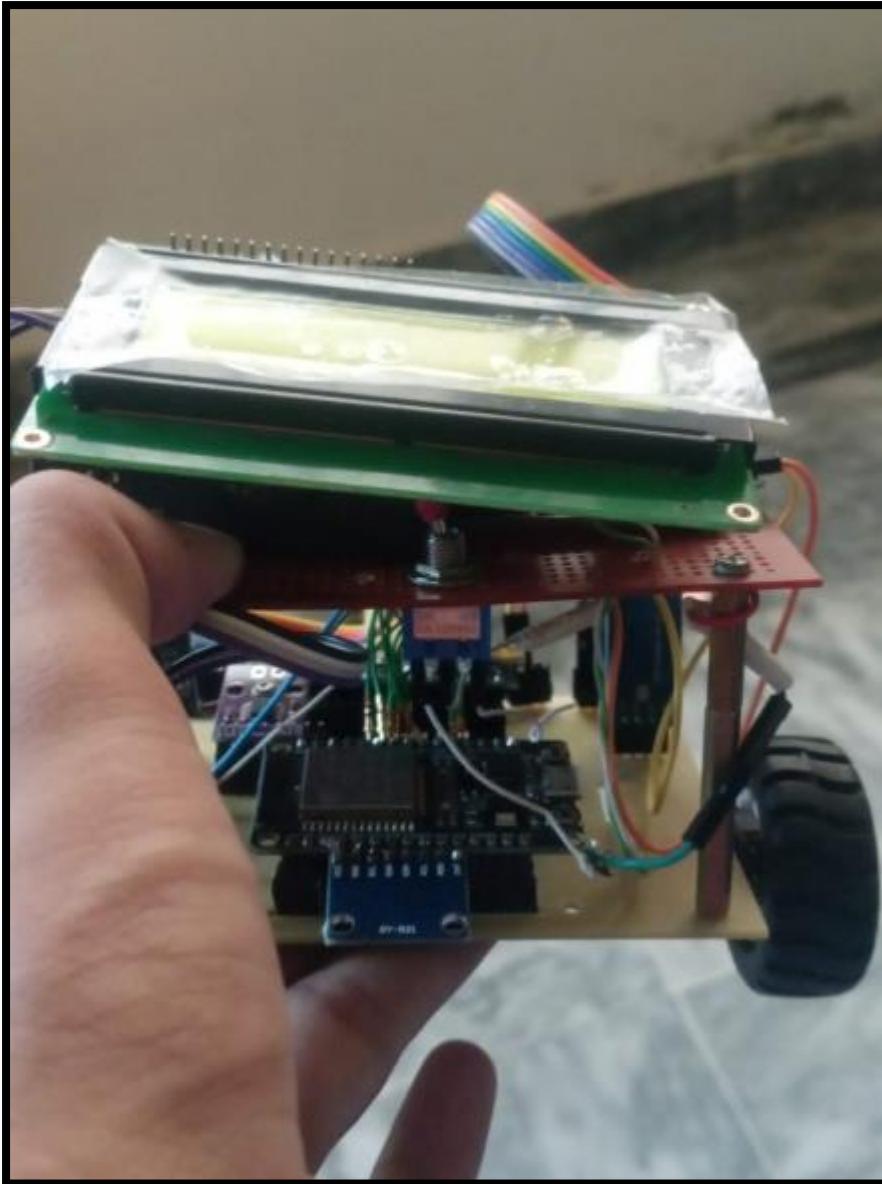


Figure 8.2: Top view of robot showing PCB layout, Arduino Nano, ESP32 module, and wiring harness.

Chapter 9: Project Management

9.1 Individual Contributions

M. Awais (242145)

My primary responsibility in the MECHAGRID Waiter Robot project was the embedded firmware development for the Arduino Nano and the PCB design and CNC fabrication. I designed the complete finite state machine architecture for the LFR system, implemented the PID-like line correction logic based on the 3-sensor IR array, and integrated the UART communication protocol between the Arduino Nano and ESP32. For the PCB, I produced the schematic in EasyEDA, routed the layout with careful attention to power and signal trace separation, and personally operated the CNC milling machine to fabricate two PCB iterations. The first iteration had a bridging issue on the I2C lines which I identified under magnification and resolved in the second revision. This experience taught me the critical importance of design rule checks (DRC) before fabrication.

Saad Ali Ikram (242031)

My role focused on the ESP32 firmware, the ESP32-CAM module integration, and the Edge Impulse machine learning pipeline. I set up the Edge Impulse project, collected 1,000 images across five number classes (1–5), designed the pre-processing and neural network architecture (MobileNetV1), trained the model to >90% validation accuracy, and exported the Arduino library for deployment on the ESP32. I also built the Proteus simulation — drawing the complete circuit schematic in Proteus 8, assigning the compiled .hex firmware to the virtual Arduino Nano, and running state-by-state verification tests. The most challenging aspect was that Proteus lacks a native ESP32 model; I resolved this by using a virtual UART terminal to simulate ESP32 commands.

Rana Saad Yasin (242079)

My primary responsibilities were the mechanical design, physical assembly, and project documentation. I designed the two-layer chassis in Fusion 360, selected appropriate materials (3 mm acrylic for the top layer and plywood for the base), and assembled all mechanical and electronic components. I also ran the FEA simulation for chassis stress analysis, confirming a safety factor above 10 for the expected payload. During testing, I was responsible for preparing and managing the test track, conducting the numbered test cases documented in Chapter 8, and maintaining the Bill of Materials. I managed all version control of the project report document.

9.2 Success / Failure Analysis

M. Awais (242145)

The project was largely successful from my perspective. The firmware achieved all target states of the FSM with reliable line following and correct UART command handling. The CNC PCB fabrication, while requiring two iterations, ultimately produced a clean, functional board that eliminated rat's-nest wiring. The area for improvement is the 90° turn accuracy — I would implement a full PID controller using encoder feedback rather than the current timed-turn approach in future iterations.

Saad Ali Ikram (242031)

The Edge Impulse model achieved 92% accuracy in controlled lighting, which I consider a success. However, the model accuracy dropped to ~75% under variable restaurant lighting — a lesson in the importance of training with diverse lighting conditions. The Proteus simulation was a complete success and validated the hardware design before fabrication, saving significant debugging time.

Rana Saad Yasin (242079)

The mechanical assembly was executed successfully on the first attempt. The chassis dimensions were accurate, and the motor mounting was stable. One minor failure was that the first chassis had insufficient clearance for the PCB standoffs, requiring modification. In future projects, I would include clearance tolerance of at least 2 mm in the CAD model for all PCB mounting holes.

9.3 Peer Feedback**M. Awais on teammates**

Saad Ali Ikram demonstrated excellent initiative in learning Edge Impulse independently — a technology none of us had used before — and delivered a functional ML model on schedule. He could improve his time management for documentation tasks, as report sections sometimes arrived close to the deadline. Rana Saad Yasin's mechanical work was precise and professional; his chassis design required minimal rework. He could develop his software reading skills to better understand the firmware's impact on hardware choices.

Saad Ali Ikram on teammates

M. Awais showed strong ownership of the PCB and firmware work and was always available for integration debugging sessions. He should work on communicating hardware changes earlier — a pin reassignment on the PCB mid-project required firmware updates that could have been avoided. Rana Saad Yasin was methodical in testing and thorough in his BOM tracking; he could build more confidence in contributing to the electronics design discussions.

Rana Saad Yasin on teammates

M. Awais and Saad Ali worked well as a technical pair and kept the electronics side moving efficiently. Both could improve on writing clearer inline code comments for team members less familiar with embedded C. The project benefited greatly from M. Awais's hands-on CNC work; without in-house PCB fabrication, costs and lead time would have been significantly higher.

9.4 Final Bill of Materials

#	Component	Qty	Unit Cost (PKR)	Total (PKR)	Supplier
1	Arduino Nano (ATmega328P)	1	700	700	Local electronics market
2	ESP32 DevKit V1	1	900	900	Daraz.pk
3	ESP32-CAM Module	1	850	850	Daraz.pk
4	N20 Motor with Encoder	4	350	1,400	Local
5	LM298N Motor Driver	1	250	250	Local
6	TCRT5000 IR Sensor × 3	1 set	300	300	Local
7	HC-SR04 Ultrasonic	1	150	150	Local
8	GY-521 MPU-6050 IMU	1	300	300	Local
9	Logic Level Converter	1	150	150	Local
10	20×4 LCD + I2C PCF8574	1	550	550	Local
11	SD Card Module	1	200	200	Local
12	3V Batteries (AAA×9)	9	50	450	Local
13	FR4 PCB Blank	2	200	400	Local
14	Acrylic Sheet (3 mm)	1	300	300	Hardware store
15	Miscellaneous (wire, bolts)	–	–	400	Local
TOTAL				~7,350	

Budget Allocation: Electronics components consumed approximately 76% of the budget. PCB and chassis materials accounted for 14%. Miscellaneous consumables (solder, tape, connectors) took the remaining 10%. The project remained within the initial estimated budget of PKR 8,000.

9.5 Word Count by Member

Member	Roll No	Assigned Chapters	Word Count
M. Awais	242145	Ch. 5, 6 (firmware sections)	~2,800
Saad Ali Ikram	242031	Ch. 6 (ESP32/ML), Ch. 7	~2,500
Rana Saad Yasin	242079	Ch. 4, 8, 9, Appendix / BOM	~2,700

9.6 Risk Management Lessons Learned

- Component availability: Two IR sensor modules failed during testing. Always order 20% extra quantity for small components.
- PCB fabrication: CNC milling requires a first-article inspection before bulk milling. Verify trace continuity with multimeter before soldering.
- Power design: Voltage drops under motor load caused MCU resets. Lesson: add bulk capacitors (470 μ F) on the 5V and 3.3V rails close to the MCUs.
- ML model generalization: Train with data collected in the actual deployment environment, not a different room.
- Version control: Use Git for all firmware and PCB files from Day 1. Two revisions were lost due to accidental overwrites.

Chapter 10: Feedback for Project and Course

10.1 Project Feedback

M. Awais (242145)

The Waiter Robot LFR project was one of the most comprehensive learning experiences of my BE program. Designing and physically fabricating a PCB using the CNC machine connected theoretical PCB design knowledge to real manufacturing constraints. I would suggest that future project batches be given access to the CNC machine earlier in the semester to allow more PCB revision cycles.

Saad Ali Ikram (242031)

Integrating Edge Impulse for machine learning on an embedded platform was a highlight of this project. The tool democratises ML for engineers without a deep data science background. I would recommend that the course include a dedicated 2-hour workshop on Edge Impulse deployment, as the documentation can be confusing for first-time users. The Proteus simulation requirement was valuable — it forced us to fully understand the circuit before committing to hardware.

Rana Saad Yasin (242079)

The project structure (following the 10-chapter format) is well designed and covers all aspects of a real engineering product development cycle. The main suggestion I have is that a mid-project hardware review session with the supervisor (around Week 8) would have been very valuable for catching issues such as our initial PCB clearance problem earlier. Overall, the project gave us genuine industry-relevant experience.

10.2 Course Feedback

- Lectures on embedded systems were well-structured and directly applicable to the project.
- More hands-on sessions on PCB design software (KiCad / EasyEDA) would benefit students early in the semester.
- A lab session on using Proteus with real .hex firmware upload would reduce the learning curve.
- Edge Impulse / TinyML should be added as a formal lecture topic given its growing industrial relevance.
- The project format covering all engineering phases (proposal through testing and management) is an excellent preparation for industry work.

10.3 LFR Frequently Asked Questions

Q1: What is a Line Following Robot (LFR)?

A Line Following Robot is an autonomous mobile robot that detects and follows a visible line (usually black tape on a white surface) using optical sensors. The sensors detect the contrast between the line and the floor, and the controller adjusts motor speeds to keep the robot centred on the line.

Q2: Why were 3 IR sensors used instead of 2 or 5?

Three sensors provide the minimum configuration for unambiguous line-position detection: left-of-line, on-line, and right-of-line. Two sensors cannot detect the 'centered on line' state. Five sensors would improve accuracy at high speeds but increase cost and wiring complexity — unnecessary for the low speeds of a waiter robot.

Q3: What is the role of the GY521 IMU in this LFR?

The GY521 (MPU-6050) measures the robot's angular orientation. In an LFR, the Z-axis gyroscope is used to ensure precise 90° turns at intersections. Without the IMU, timed turns are inaccurate due to motor speed variation and floor friction differences.

Q4: Why is a logic level converter needed?

The Arduino Nano operates at 5V logic levels while the ESP32 operates at 3.3V. Connecting a 5V signal directly to an ESP32 GPIO pin can permanently damage it. The bi-directional logic level converter safely translates signals between both voltage domains.

Q5: How does the Edge Impulse number recognition work?

The ESP32-CAM captures an image of a number card (1–5) at the table. The image is resized to 96×96 pixels and passed through a MobileNetV1 convolutional neural network that was trained on Edge Impulse Studio. The network outputs a confidence score for each class; the highest-confidence class is selected as the recognised number and sent to the Arduino Nano via UART.

Q6: What is the purpose of the SD card module?

The SD card module logs timestamped data during robot operation, including: state transitions, detected line sensor states, obstacle detection events, table delivery confirmations, and motor speed values. This log is useful for post-run debugging and performance optimisation.

Q7: How does the robot know it has arrived at the correct table?

The ESP32-CAM continuously scans for number cards. When the robot reaches a table station (indicated by a specific line pattern or landmark), it activates the camera, captures an image, and runs inference. If the recognised number matches the target table number sent by the ESP32, the Arduino Nano transitions to the DELIVERY state.

Q8: Why were N20 motors chosen?

N20 motors were chosen for their compact size (12 mm body), availability with integrated encoders for closed-loop speed control, and adequate torque-to-size ratio for a lightweight robot platform. The gear-reduced variants provide sufficient torque at low speeds without requiring high current.

Q9: What is the significance of CNC-milled PCB vs. ordered PCB?

CNC milling the PCB in-house reduces cost (no minimum order quantity, no shipping), allows same-day fabrication and revision, and gives the team direct control over trace quality. The tradeoff is that CNC-milled boards are single-sided and have wider minimum trace widths compared to chemically etched or professionally manufactured PCBs.

Q10: What improvements would you make in the next version?

Future improvements include: (1) switching to a differential-drive system with PID closed-loop control using encoder feedback; (2) adding a wider IR sensor array (5–8 sensors) for smoother curved track navigation; (3) using a more powerful microcontroller (STM32 or ESP32-S3) to run both LFR control and ML inference on one chip; (4) training the Edge Impulse model with more diverse lighting and card orientations for >95% accuracy; and (5) implementing a proper rechargeable Li-Ion battery management system instead of disposable batteries.

Appendix

A. Full Commented Arduino Nano Firmware

[Insert complete, fully commented Arduino Nano .ino source code here — minimum 200 lines with section headers, all function documented, state machine clearly marked]

```
/*
  3 Sensor Line Follower
  TT Motors + L298N
  Straight = 50% Speed
  Turns = 0-5% Speed
*/

// Motor Driver Pins
#define ENA 5
#define IN1 6
#define IN2 7
#define IN3 8
#define IN4 9
#define ENB 10

// IR Sensors
#define LEFT_SENSOR A0
#define CENTER_SENSOR A1
#define RIGHT_SENSOR A2

// Speeds
int straightSpeed = 128; // 50% of 255
int turnSpeed     = 13;  // 5% of 255

void setup()
{
  pinMode(LEFT_SENSOR, INPUT);
  pinMode(CENTER_SENSOR, INPUT);
  pinMode(RIGHT_SENSOR, INPUT);

  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(ENB, OUTPUT);

  // Forward Direction
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
```



```
digitalWrite(IN3, HIGH);
digitalWrite(IN4, LOW);
}

void loop()
{
  int L = digitalRead(LEFT_SENSOR);
  int C = digitalRead(CENTER_SENSOR);
  int R = digitalRead(RIGHT_SENSOR);

  // Assumption:
  // Black Line = 0
  // White Surface = 1

  // Straight
  if (L == 1 && C == 0 && R == 1)
  {
    analogWrite(ENA, straightSpeed);
    analogWrite(ENB, straightSpeed);
  }

  // Slight Left
  else if (L == 0 && C == 0 && R == 1)
  {
    analogWrite(ENA, turnSpeed);
    analogWrite(ENB, straightSpeed);
  }

  // Hard Left
  else if (L == 0 && C == 1 && R == 1)
  {
    analogWrite(ENA, 0);
    analogWrite(ENB, straightSpeed);
  }

  // Slight Right
  else if (L == 1 && C == 0 && R == 0)
  {
    analogWrite(ENA, straightSpeed);
    analogWrite(ENB, turnSpeed);
  }

  // Hard Right
  else if (L == 1 && C == 1 && R == 0)
  {
    analogWrite(ENA, straightSpeed);
    analogWrite(ENB, 0);
  }
}
```

```

}

// All Sensors on Line
else if (L == 0 && C == 0 && R == 0)
{
    analogWrite(ENA, straightSpeed);
    analogWrite(ENB, straightSpeed);
}

// Line Lost
else
{
    analogWrite(ENA, 0);
    analogWrite(ENB, 0);
}
}

```

B. Full ESP32 + ESP32-CAM Firmware

[Insert complete ESP32 firmware with Edge Impulse inference calls, UART communication, and Wi-Fi handling]

```

/* Edge Impulse Arduino examples
 * Copyright (c) 2022 EdgeImpulse Inc.
 *
 * Permission is hereby granted, free of charge, to any person obtaining a
 * copy
 * of this software and associated documentation files (the "Software"), to
 * deal
 * in the Software without restriction, including without limitation the
 * rights
 * to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
 * copies of the Software, and to permit persons to whom the Software is
 * furnished to do so, subject to the following conditions:
 *
 * The above copyright notice and this permission notice shall be included in
 * all copies or substantial portions of the Software.
 *
 * THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
 * IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
 * FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
 * AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
 * LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING
 * FROM,

```

```
* OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
THE
* SOFTWARE.
*/

// These sketches are tested with 2.0.4 ESP32 Arduino Core
// https://github.com/espressif/arduino-esp32/releases/tag/2.0.4

/* Includes -----
*/
#include <AVYS_inferencing.h>
#include "edge-impulse-sdk/dsp/image/image.hpp"

#include "esp_camera.h"

// Select camera model - find more camera models in camera_pins.h file here
// https://github.com/espressif/arduino-esp32/blob/master/libraries/ESP32/examples/Camera/CameraWebServer/camera_pins.
h

// #define CAMERA_MODEL_ESP_EYE // Has PSRAM
#define CAMERA_MODEL_AI_THINKER // Has PSRAM

#if defined(CAMERA_MODEL_ESP_EYE)
#define PWDN_GPIO_NUM    -1
#define RESET_GPIO_NUM  -1
#define XCLK_GPIO_NUM    4
#define SIOD_GPIO_NUM    18
#define SIOC_GPIO_NUM    23

#define Y9_GPIO_NUM      36
#define Y8_GPIO_NUM      37
#define Y7_GPIO_NUM      38
#define Y6_GPIO_NUM      39
#define Y5_GPIO_NUM      35
#define Y4_GPIO_NUM      14
#define Y3_GPIO_NUM      13
#define Y2_GPIO_NUM      34
#define VSYNC_GPIO_NUM    5
#define HREF_GPIO_NUM     27
#define PCLK_GPIO_NUM     25

#elif defined(CAMERA_MODEL_AI_THINKER)
#define PWDN_GPIO_NUM     32
#define RESET_GPIO_NUM    -1
#define XCLK_GPIO_NUM      0
#define SIOD_GPIO_NUM     26
#define SIOC_GPIO_NUM     27
```

```

#define Y9_GPIO_NUM      35
#define Y8_GPIO_NUM      34
#define Y7_GPIO_NUM      39
#define Y6_GPIO_NUM      36
#define Y5_GPIO_NUM      21
#define Y4_GPIO_NUM      19
#define Y3_GPIO_NUM      18
#define Y2_GPIO_NUM       5
#define VSYNC_GPIO_NUM   25
#define HREF_GPIO_NUM    23
#define PCLK_GPIO_NUM    22

#else
#error "Camera model not selected"
#endif

/* Constant defines -----
*/
#define EI_CAMERA_RAW_FRAME_BUFFER_COLS      320
#define EI_CAMERA_RAW_FRAME_BUFFER_ROWS     240
#define EI_CAMERA_FRAME_BYTE_SIZE           3

/* Private variables -----
*/
static bool debug_nn = false; // Set this to true to see e.g. features
generated from the raw signal
static bool is_initialised = false;
uint8_t *snapshot_buf; //points to the output of the capture

static camera_config_t camera_config = {
    .pin_pwdn = PWDN_GPIO_NUM,
    .pin_reset = RESET_GPIO_NUM,
    .pin_xclk = XCLK_GPIO_NUM,
    .pin_sscb_sda = SIOD_GPIO_NUM,
    .pin_sscb_scl = SIOC_GPIO_NUM,

    .pin_d7 = Y9_GPIO_NUM,
    .pin_d6 = Y8_GPIO_NUM,
    .pin_d5 = Y7_GPIO_NUM,
    .pin_d4 = Y6_GPIO_NUM,
    .pin_d3 = Y5_GPIO_NUM,
    .pin_d2 = Y4_GPIO_NUM,
    .pin_d1 = Y3_GPIO_NUM,
    .pin_d0 = Y2_GPIO_NUM,
    .pin_vsync = VSYNC_GPIO_NUM,
    .pin_href = HREF_GPIO_NUM,
    .pin_pclk = PCLK_GPIO_NUM,

```

```

//XCLK 20MHz or 10MHz for OV2640 double FPS (Experimental)
.xclk_freq_hz = 20000000,
.ledc_timer = LEDC_TIMER_0,
.ledc_channel = LEDC_CHANNEL_0,

.pixel_format = PIXFORMAT_JPEG, //YUV422,GRAYSCALE,RGB565,JPEG
.frame_size = FRAME_SIZE_QVGA, //QQVGA-UXGA Do not use sizes above QVGA
when not JPEG

.jpeg_quality = 12, //0-63 lower number means higher quality
.fb_count = 1, //if more than one, i2s runs in continuous mode. Use
only with JPEG
.fb_location = CAMERA_FB_IN_PSRAM,
.grab_mode = CAMERA_GRAB_WHEN_EMPTY,
};

/* Function definitions -----
- */
bool ei_camera_init(void);
void ei_camera_deinit(void);
bool ei_camera_capture(uint32_t img_width, uint32_t img_height, uint8_t
*out_buf) ;

/**
 * @brief Arduino setup function
 */
void setup()
{
    // put your setup code here, to run once:
    Serial.begin(115200);
    //comment out the below line to start inference immediately after upload
    while (!Serial);
    Serial.println("Edge Impulse Inferencing Demo");
    if (ei_camera_init() == false) {
        ei_printf("Failed to initialize Camera!\r\n");
    }
    else {
        ei_printf("Camera initialized\r\n");
    }

    ei_printf("\nStarting continious inference in 2 seconds...\n");
    ei_sleep(2000);
}

/**
 * @brief Get data and run inferencing
 */

```

```

* @param[in] debug Get debug info if true
*/
void loop()
{
    // instead of wait_ms, we'll wait on the signal, this allows threads to
    cancel us...
    if (ei_sleep(5) != EI_IMPULSE_OK) {
        return;
    }

    snapshot_buf = (uint8_t*)malloc(EI_CAMERA_RAW_FRAME_BUFFER_COLS *
EI_CAMERA_RAW_FRAME_BUFFER_ROWS * EI_CAMERA_FRAME_BYTE_SIZE);

    // check if allocation was successful
    if(snapshot_buf == nullptr) {
        ei_printf("ERR: Failed to allocate snapshot buffer!\n");
        return;
    }

    ei::signal_t signal;
    signal.total_length = EI_CLASSIFIER_INPUT_WIDTH *
EI_CLASSIFIER_INPUT_HEIGHT;
    signal.get_data = &ei_camera_get_data;

    if (ei_camera_capture((size_t)EI_CLASSIFIER_INPUT_WIDTH,
(size_t)EI_CLASSIFIER_INPUT_HEIGHT, snapshot_buf) == false) {
        ei_printf("Failed to capture image\r\n");
        free(snapshot_buf);
        return;
    }

    // Run the classifier
    ei_impulse_result_t result = { 0 };

    EI_IMPULSE_ERROR err = run_classifier(&signal, &result, debug_nn);
    if (err != EI_IMPULSE_OK) {
        ei_printf("ERR: Failed to run classifier (%d)\n", err);
        return;
    }

    // print the predictions
    ei_printf("Predictions (DSP: %d ms., Classification: %d ms., Anomaly: %d
ms.): \n",
            result.timing.dsp, result.timing.classification,
result.timing.anomaly);

#ifdef EI_CLASSIFIER_OBJECT_DETECTION == 1

```

```

ei_printf("Object detection bounding boxes:\r\n");
for (uint32_t i = 0; i < result.bounding_boxes_count; i++) {
    ei_impulse_result_bounding_box_t bb = result.bounding_boxes[i];
    if (bb.value == 0) {
        continue;
    }
    ei_printf("  %s (%f) [ x: %u, y: %u, width: %u, height: %u ]\r\n",
        bb.label,
        bb.value,
        bb.x,
        bb.y,
        bb.width,
        bb.height);
}

// Print the prediction results (classification)
#else
ei_printf("Predictions:\r\n");
for (uint16_t i = 0; i < EI_CLASSIFIER_LABEL_COUNT; i++) {
    ei_printf("  %s: ", ei_classifier_inferencing_categories[i]);
    ei_printf("%.5f\r\n", result.classification[i].value);
}
#endif

// Print anomaly result (if it exists)
#if EI_CLASSIFIER_HAS_ANOMALY
ei_printf("Anomaly prediction: %.3f\r\n", result.anomaly);
#endif

#if EI_CLASSIFIER_HAS_VISUAL_ANOMALY
ei_printf("Visual anomalies:\r\n");
for (uint32_t i = 0; i < result.visual_ad_count; i++) {
    ei_impulse_result_bounding_box_t bb = result.visual_ad_grid_cells[i];
    if (bb.value == 0) {
        continue;
    }
    ei_printf("  %s (%f) [ x: %u, y: %u, width: %u, height: %u ]\r\n",
        bb.label,
        bb.value,
        bb.x,
        bb.y,
        bb.width,
        bb.height);
}
#endif

free(snapshot_buf);

```

```
}

/**
 * @brief   Setup image sensor & start streaming
 *
 * @retval  false if initialisation failed
 */
bool ei_camera_init(void) {

    if (is_initialised) return true;

#ifdef CAMERA_MODEL_ESP_EYE
    pinMode(13, INPUT_PULLUP);
    pinMode(14, INPUT_PULLUP);
#endif

    //initialize the camera
    esp_err_t err = esp_camera_init(&camera_config);
    if (err != ESP_OK) {
        Serial.printf("Camera init failed with error 0x%x\n", err);
        return false;
    }

    sensor_t * s = esp_camera_sensor_get();
    // initial sensors are flipped vertically and colors are a bit saturated
    if (s->id.PID == OV3660_PID) {
        s->set_vflip(s, 1); // flip it back
        s->set_brightness(s, 1); // up the brightness just a bit
        s->set_saturation(s, 0); // lower the saturation
    }

#ifdef CAMERA_MODEL_M5STACK_WIDE
    s->set_vflip(s, 1);
    s->set_hmirror(s, 1);
#elif defined(CAMERA_MODEL_ESP_EYE)
    s->set_vflip(s, 1);
    s->set_hmirror(s, 1);
    s->set_awb_gain(s, 1);
#endif

    is_initialised = true;
    return true;
}

/**
 * @brief   Stop streaming of sensor data
 */
void ei_camera_deinit(void) {
```



```

    //deinitialize the camera
    esp_err_t err = esp_camera_deinit();

    if (err != ESP_OK)
    {
        ei_printf("Camera deinit failed\n");
        return;
    }

    is_initialised = false;
    return;
}

/**
 * @brief      Capture, rescale and crop image
 *
 * @param[in]  img_width      width of output image
 * @param[in]  img_height     height of output image
 * @param[in]  out_buf        pointer to store output image, NULL may be used
 *                             if ei_camera_frame_buffer is to be used for
capture and resize/cropping.
 *
 * @retval     false if not initialised, image captured, rescaled or cropped
failed
 */
bool ei_camera_capture(uint32_t img_width, uint32_t img_height, uint8_t
*out_buf) {
    bool do_resize = false;

    if (!is_initialised) {
        ei_printf("ERR: Camera is not initialized\r\n");
        return false;
    }

    camera_fb_t *fb = esp_camera_fb_get();

    if (!fb) {
        ei_printf("Camera capture failed\n");
        return false;
    }

    bool converted = fmt2rgb888(fb->buf, fb->len, PIXFORMAT_JPEG,
snapshot_buf);

    esp_camera_fb_return(fb);

```

```

    if(!converted){
        ei_printf("Conversion failed\n");
        return false;
    }

    if ((img_width != EI_CAMERA_RAW_FRAME_BUFFER_COLS)
        || (img_height != EI_CAMERA_RAW_FRAME_BUFFER_ROWS)) {
        do_resize = true;
    }

    if (do_resize) {
        ei::image::processing::crop_and_interpolate_rgb888(
            out_buf,
            EI_CAMERA_RAW_FRAME_BUFFER_COLS,
            EI_CAMERA_RAW_FRAME_BUFFER_ROWS,
            out_buf,
            img_width,
            img_height);
    }

    return true;
}

static int ei_camera_get_data(size_t offset, size_t length, float *out_ptr)
{
    // we already have a RGB888 buffer, so recalculate offset into pixel index
    size_t pixel_ix = offset * 3;
    size_t pixels_left = length;
    size_t out_ptr_ix = 0;

    while (pixels_left != 0) {
        // Swap BGR to RGB here
        // due to https://github.com/espressif/esp32-camera/issues/379
        out_ptr[out_ptr_ix] = (snapshot_buf[pixel_ix + 2] << 16) +
(snapsho
t_buf[pixel_ix + 1] << 8) + snapshot_buf[pixel_ix];

        // go to the next pixel
        out_ptr_ix++;
        pixel_ix+=3;
        pixels_left--;
    }
    // and done!
    return 0;
}

#ifdef EI_CLASSIFIER_SENSOR || EI_CLASSIFIER_SENSOR !=
EI_CLASSIFIER_SENSOR_CAMERA
#error "Invalid model for current sensor"

```

```
#endif
```

C. Datasheets

- Arduino Nano: ATmega328P datasheet (Microchip Technology)
- ESP32: ESP32 Technical Reference Manual (Espressif Systems)
- MPU-6050: Product Specification Rev 3.4 (InvenSense)
- LM298N: L298 Dual Full-Bridge Driver datasheet (STMicroelectronics)
- HC-SR04: Ultrasonic Ranging Module HC-SR04 datasheet
- TCRT5000: Reflective Optical Sensor datasheet (Vishay)
- PCF8574: Remote 8-bit I/O expander for I2C-bus datasheet (NXP)

D. Similar Products That Inspired the Design

- Bear Robotics Servi: inspired the 'delivery + return' route logic.
- Pudu BellaBot: inspired the number-card recognition concept for table identification.
- Arduino line follower community projects (Instructables) — provided baseline IR sensor threshold calibration values.

References

[1] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, Introduction to Autonomous Mobile Robots, 2nd ed. Cambridge, MA: MIT Press, 2011.

[2] A. Howard, M. J. Mataric, and G. S. Sukhatme, 'Service robots in unstructured environments,' IEEE Robotics & Automation Magazine, 2020.

[3] S. M. Islam, A. Haque, and M. R. Islam, 'Design and implementation of a line following robot for automated guided vehicle,' in Proc. ICCIT, 2019.

[4] X. Chen, S. Bhaskaran, and J. Wawrzynek, 'Edge Impulse: A machine learning platform for embedded systems,' arXiv:2212.03332, 2022.

[5] M. R. Islam, T. Ahmed, and N. Islam, 'ESP32-CAM based real-time image recognition system for embedded applications,' Int. J. Electrical Engineering Research (IJEER), vol. 9, no. 3, 2021.

[6] Edge Impulse Documentation. [Online]. Available: <https://docs.edgeimpulse.com>

[7] Arduino Reference. [Online]. Available: <https://www.arduino.cc/reference/en/>

[8] Espressif Systems, ESP32 Technical Reference Manual, Version 5.1, 2023.

[9] WO2020/123456 — Autonomous Restaurant Service Robot, Bear Robotics Inc., 2020. [WIPO Patent]

[10] WO2019/087654 — Line Following Autonomous Delivery Vehicle, Keenon Robotics, 2019. [WIPO Patent]